

The Development of Risk Attitudes and their Cultural Transmission

Alejandro Pérez Velilla¹, Bret Beheim², Paul E. Smaldino^{1,3}

¹ University of California, Merced

² Max Planck Institute for Evolutionary Anthropology

³ Santa Fe Institute

Abstract

We use cultural evolutionary models to examine how individual experiences and culturally-inherited information jointly shape risk attitudes under environmental uncertainty. We find that learning processes not only generate plausible variation in risk attitudes, but also that conservative learning strategies—emphasizing the preservation of generational knowledge—excel in high-risk environments, promoting stable wealth accumulation and long-term survival but limiting asset growth as conditions improve. In contrast, exploratory learning strategies—leveraging risk-free juvenile exploration and peer influence—foster risk-tolerant attitudes that thrive in affluent, low-risk settings where wealth buffers and social safety nets reduce the costs of miscalculations. Introducing economic stratification to the model reveals how wealth disparities and inter-class interactions reinforce these patterns, exacerbating differences in learning strategies and risk-taking behaviors, and perpetuating socioeconomic inequalities through the cultural inertia of excessive risk aversion. By uniting developmental, social, and evolutionary perspectives, our framework provides a novel lens on the cultural evolution of risk attitudes and their broader societal implications.

Keywords: uncertainty; social learning; cultural evolution; poverty traps; parochialism; decision making

1 Introduction

Research on risk attitudes and on social learning strategies have, for the most part, developed independently. This is unfortunate due to the importance of social learning in helping organisms navigate environmental uncertainty (Boyd and Richerson, 1985; Laland, 2004; Kendal et al., 2018) and, more generally, in explaining the extent of human psychological and behavioral variation across the globe (Chudek and Henrich, 2011; Gelfand et al., 2024). In this paper, we integrate these topics and focus on the cultural acquisition of risk attitudes: psychological preferences that govern how individuals use information to make decisions in uncertain conditions.

Understanding the origins of risk attitudes is central to many fields, from economics and psychology to anthropology and evolutionary biology, where risk-relevant behavior is often viewed from a life-history lens (Nettle and Frankenhuys, 2020). Risk attitudes are critical for understanding human behavior, particularly in domains where decisions resemble gambles, such as economic activities, educational pursuits, mate selection, and parental investment. Acquiring optimal risk attitudes—those that yield the best outcomes for a given environment—might be straightforward in an environment that is fully predictable or where all risks are accurately known. Real-world conditions are inherently

uncertain in the Knightian sense (Knight, 1921). In risky settings with the potential for catastrophic loss, moreover, caution often becomes essential. The challenge of optimizing risk attitudes is therefore difficult, if not impossible, for an individual to solve without help.

Economic approaches often assume that individuals are able to rationally maximize expected utility, allowing for the quantification of risk attitudes across different scenarios (Harrison and Rutström, 2008), though evolutionary-oriented economic models have also posited that some degree of risk aversion is adaptive, reflecting evolved traits for navigating uncertain environments where rational decision-making is key to survival (Robson and Samuelson, 2019; Robson and Orr, 2021; Robson and Samuelson, 2022; Alger, 2023). Psychological research tends to emphasize cognitive and emotional factors in risk perception, viewing childhood and adolescence as critical periods for refining risk-related behaviors in response to adversity (Gopnik, 2020; Frankenhuis and Gopnik, 2023; Xu et al., 2023), while work in evolutionary biology, behavioral ecology, and anthropology highlights how aversion to risk can evolve under varying environmental conditions (Levy, 2010; Zhang et al., 2014; Schreiber, 2015; Mallpress et al., 2015; Netzer et al., 2022). Despite the existence of these rich perspectives, they remain largely disconnected. A cultural evolutionary viewpoint—bridging economic, cognitive, social, and evolutionary sciences—offers a framework for integrating these insights and generating new testable hypotheses.

Here, we develop a model that clarifies how risk attitudes emerge from the interplay of evolved predispositions for navigating risk-reward trade-offs, individual life experiences, and cultural transmission. By doing so, we unify previously isolated strands of research and explain variation in risk attitudes across individuals, groups, and socioeconomic contexts through individual and social learning mechanisms. We answer calls to incorporate culture into models of risk behavior (Weber and Hsee, 1999; Bowles, 1998) by proposing that, through different forms of learning, risk attitudes can change both within and across generations at rates faster than those permitted by genetic evolution—akin to a “Lamarckian” cultural process in which accumulated environmental knowledge is passed on to future cohorts.

How do individuals discern whether their environment rewards investments or punishes them with ruinous losses? Trial-and-error learning is possible, but costly—especially when information is noisy or slow to accumulate. Childhood may act as a protective period that permits exploration and peer learning with minimal immediate risks (Gopnik, 2020; Frankenhuis and Gopnik, 2023; Xu et al., 2023) at the cost of omitting important information about those risks. As individuals mature, additional learning occurs through direct experience and from observing others’ successes, failures and catastrophes. When environmental risk changes little between generations, vertical or oblique transmission of established practices from elders can be a highly effective means of stabilizing near-optimal risk attitudes (Boyd and Richerson, 1985; Deffner and McElreath, 2022; Turner et al., 2023), avoiding optimistic (too risk tolerant) or pessimistic (too risk averse) attitudes.

Environments vary widely, however, and not all individuals within an environment face identical conditions. Wealth, institutional stability, and social safety nets can strongly influence the consequences of risk-taking. Research suggests that risk attitudes are not fixed; they develop over the lifespan and evolve across generations, shaped by changing conditions, cultural norms, and individual adversity (Amir et al., 2020; Xu et al., 2023). For example, younger individuals often show higher risk tolerance, which may decline with age (Paulsen et al., 2012; Leonard and Sommerville, 2024). Early-life adversity can accelerate the onset of risk aversion (Gopnik, 2020; Frankenhuis and Gopnik, 2023; Xu et al., 2023), and shocking or traumatic experiences can create long-lasting shifts toward heightened caution (Moya, 2018; Kim and Lee, 2014; Augsburger and Elbert, 2017). Moreover, while individual variation in risk aversion is often unexplored in models of optimal risk-taking, empirical work suggests that different socioeconomic strata can exhibit complex risk attitude distributions with distinct variance signatures (de Courson et al., 2024). Environmental conditions can

also change with time, such that strategies that previously worked well are no longer effective.

Taking these lines of research into account, we propose that risk attitudes emerge from a combination of evolutionary and developmental processes, as evolved predispositions interact with experiential and cultural inputs. Four key factors shape this process: 1) a baseline level of risk aversion, shaped by natural selection; 2) risk-free environmental sampling during early developmental periods, reinforced by horizontal learning from peer interactions (exploratory learning); 3) vertical and oblique cultural transmission of risk-related knowledge from previous generations (conservative learning); and 4) lifetime adjustments in response to social information that indicates that one’s current risk attitudes are overly optimistic (sensitivity to peer losses).

Cultural selection acts on each of these factors, guiding the lifelong acquisition of risk attitudes. Our model reveals how the cultural evolution of different strategies for learning about risk-taking behavior—ranging from conservative to exploratory—yield distinct cultural equilibria in risk attitudes. Conservative learning strategies, characterized by the strong influence of information inherited from elders, thrive in high-risk settings but can delay adaptation when conditions improve. Exploratory learning strategies, which rely more on immediate environmental feedback and peer reinforcement of said information, are better suited to improving conditions, but may lead to ruin if misapplied in risky environments. Generally, we find that conservative learning strategies outperform exploratory ones, except under low-risk, high-wealth conditions where the insulation provided by ample resource buffers encourages boldness and diversification in information sources, as well as riskier and potentially maladaptive learning strategies such as payoff-biased learning (i.e., preferentially learning strategies for managing risk from wealthy individuals).

We also examine the effects of socioeconomic stratification by allowing group-specific differences in environmental risk and wealth buffers. In such scenarios, wealthy groups can afford more risk-taking and may adopt payoff bias as a parochial strategy, reinforcing their advantage and cultural distinctiveness. In contrast, disadvantaged groups, facing harsher conditions, remain trapped in conservative, risk-averse equilibria, often resorting to parochial learning to avoid ill-fitting information gleaned from wealthier out-groups. This leads to entrenched socioeconomic disparities and cultural inertia in risk attitudes.

We begin with a simple investment dynamic in which agents must gamble to increase their wealth and face a risk-reward trade-off: the more risk they take, the higher their potential return on investment but also the greater their risk of ruin increases. To this we add developmental stages in which agents use individual and peer learning as ways to acquire information and estimate their environmental conditions, allowing them to choose risk attitudes attuned to their present conditions, as well as to quickly adjust those attitudes in response to witnessing the ruin of other individuals. We then add a layer of cultural inheritance, in which agents can use information from previous generations to further hone their environmental estimates and risk attitudes. We will show that inter-generational transmission greatly improves adaptive performance, even in environments undergoing fast and significant changes. We show that learning dynamics not only generate considerable individual-level variation in risk attitudes, but also that, at the population level, this variation is highly dependent on the degree of risk the population is subject to. Finally, we explore populations that are stratified by class and study the emergence of class differences in risk-learning strategies and subsequent class differences in responses to environmental change.

In general, our focus shifts from identifying optimal risk attitudes to exploring the optimal strategies for *learning about risk*. That is, we ask how agents use different learning strategies to acquire adaptive behaviors over their lifespans, given environmental variability. We show that environmental uncertainty and wealth buffers lead to distinct *optimal learning strategies*, which are used by individuals to learn about the risk in their environments. A question that might arise to readers is: why explore the evolution of learning strategies? Why don’t agents just evolve their risk attitudes

instead of acquiring them through a complex (and error-prone) process of individual and social learning? Our answer is that the flexibility of learning is effective at dealing with environmental change, both within the lifetime and across generations, which is typically too rapid and variable across environments for natural selection to track (Rogers, 1988; Boyd et al., 2011).

Humans in the real world do not come pre-loaded with fully-realized risk attitudes. Rather, they acquire risk attitudes by integrating environmental social information to better attune their attitudes to the world in which they live. This has advantages: relying on learning means that sudden changes in individual conditions and environmental risk are less likely to leave individuals completely ill-adapted to their new environments, while leaving open the possibility of inheriting tried-and-true risk attitudes in periods of stability via cultural transmission. The flexibility afforded by the ability to use information from previous generations alongside that from within-generation sources makes for fast adaptation, resulting in a two-stage “Lamarckian” cultural evolutionary process. Optimization occurs both through the continuous acquisition of information throughout the lifetime, as well as across generations by learning from elders. Cultural evolution can further refine the extent to which the different sources of information are weighted, so that risk attitudes are acquired in the best way possible.

By studying this multi-level process of developmental and cultural adaptation, we obtain several results that explain observed empirical patterns. Our analysis offers a unified explanation for observed differences in risk attitudes across developmental stages, socioeconomic strata, environmental conditions, and cultural contexts. First, individuals tend to develop pessimistic risk attitudes at lower wealth buffers, and optimistic risk attitudes at higher wealth buffers. The vagaries of learning outcomes make it so that acquiring risk attitudes that fit environmental conditions perfectly is a rare outcome, and considerable population variation in risk attitudes is an inevitable outcome of the cultural evolutionary process. Given the distinct environments that people at the lower and higher extremes of the wealth distribution face, cultures of caution and risk aversion will tend to be promoted by the former, while cultures of risk tolerance will generally be promoted by the latter.

Second, when individuals rely primarily on juvenile exploration and peer learning, they tend to adopt overly-optimistic risk attitudes during their juvenile periods, which are often suboptimal and even be catastrophic once individuals enter adulthood. Sensitivity to peer ruin therefore evolves as a critical psychological mechanism for balancing risk-taking and caution by increasing pessimism, but can also leave agents prone to extreme risk aversion (“trauma”). Since a high sensitivity to peer ruin is favored in very adverse environments, individuals are able to adjust their beliefs and attitudes quickly, but also run the risk of observing too much ruin, leaving a pessimistic imprint that can be passed down to new generations.

Third, incorporating elder influence greatly enhances agents’ ability to align risk attitudes with environmental demands. By leveraging inter-generational knowledge, populations achieve closer approximations of optimal risk-taking, even under changing environmental conditions. However, both sensitivity to peer ruin and reliance on elder influence diminish when environments become more favorable and when wealth buffers are substantial, allowing younger cohorts to benefit from more diversified and exploratory learning strategies and the higher amounts of risk tolerance they afford. In cultural evolutionary terms, higher wealth buffering weakens selection for any single source of information, allowing individuals to be more “playful” with the way they learn about the world without having to worry about high costs of getting it wrong.

Fourth, individuals in both deprived and wealthy conditions start off optimistic and become more pessimistic throughout their lifetimes, although there are intermediate conditions in which they can start off highly pessimistic and grow marginally more confident as time goes by. In deprived settings, even small influences from juvenile learning and peer influence can lead to maladaptively optimistic risk attitudes, which are later adjusted through sensitivity to peer ruin. In wealthy settings, the

diminished reliance on elder influence has a similar effect, although individuals in these wealth brackets do not adjust their risk attitudes over the lifespan nearly as much. In intermediate settings, individuals conservatively inherit information from elders that ensures they start life cautiously pessimistic, allowing for modest increases in risk tolerance through life experience.

Fifth, under conditions of severe deprivation, populations can develop a dual risk culture, in which most individuals become highly conservative in their learning and pessimistic in their risk attitudes, while a small minority adopt a maladaptive payoff bias that fosters extreme optimism (i.e., high risk tolerance), driven by rare outlier successes that disproportionately shape cultural norms. This pattern mirrors empirical findings of heightened risk-taking in marginalized groups despite widespread caution de Courson et al. (2024), and is a candidate cause for explaining, through cultural evolutionary processes, why extreme high-risk behaviors, such as criminal behavior, may appear disproportionately (although still rarely) in materially-deprived populations. In these settings, while most of those who learn from high payoff individuals (and thus end up taking large risks) face almost certain ruin, a very small number “luck out” and experience significant wealth growth. This growth can be massive relative to most non-ruined individuals in deprived settings, who use non-payoff biased conservative learning from elders, ensuring survival at the cost of minimal growth. This provides a dangerous allure for future generations of learners, who observe a highly-skewed wealth distribution among their elders. As a result, while most individuals in the next generation choose the route of caution, a substantive minority adopts payoff bias and learns from the reckless, perpetuating the cycle.

Finally, when structural inequality is introduced, learning strategies diverge along class lines. Wealthy groups, who adopt more exploratory learning (leading to greater risk tolerance), also adopt payoff bias in higher numbers as a parochial strategy, reinforcing their advantage. On the other hand, disadvantaged groups become even more reliant on conservative strategies and evolve more parochial learning cultures (in which information from outgroup individuals is actively disregarded). This divergence in learning strategies can drive the maintenance, and even further exacerbation, of inequality in risk attitudes and wealth.

Our framework provides a coherent explanation of a wide range of observed phenomena in human risk behavior, linking economic conditions, cultural processes, life history, and social learning strategies. The framework further predicts relationships between wealth, learning strategies, and risk attitudes that can guide subsequent empirical research. We offer a unified mechanistic explanation for juvenile optimism, the increase of pessimism throughout the lifetime, the transmission of acquired pessimism to new generations (“generational trauma”), and the relationship of these processes to wealth, environmental adversity, and structural inequalities. In the following sections, we detail our model, present the results of our analyses, and discuss the implications for understanding the cultural evolution of risk attitudes.

2 The model

2.1 Optimal gambling strategies for risky environments

We begin by describing a simple model of optimal gambling, upon which we will then introduce risk of ruin, development, learning, population structure, and cultural evolutionary dynamics. Numerical results throughout the paper are obtained using scripts written in the Julia programming language (Bezanson et al., 2012), with individual-based simulations assisted by the Agents.jl package (Datseris et al., 2022). All simulation code and data, as well as a Pluto notebook (JE and CT, 2021) that can generate the plots in the paper can be found at <https://github.com/datadreamscorp/PessimisticLearning>.

Wealth growth dynamics

We start with a single-asset risky investment dynamic (Kelly, 1956), representing investment decisions by agents who can opt into a multiplicative wealth growth scheme (i.e. the stock market). An agent starting with an initial wealth V_0 chooses a stake $s \in [0, 1]$, representing the fraction of their wealth they will bet, and then proceeds to throw a biased coin every time period (for a total of T periods), so that each attempt results in a success with probability u and a failure with probability $1 - u$. We refer to an agent's chosen stake as their *risk attitude*, as it describes their confidence that their gamble will pay off and implies the fraction of their wealth they are willing to forego in case of failure.

If a gamble in a particular time period is successful, the player increases their wealth by a factor $1 + s$, otherwise their wealth decreases by a factor $1 - s$. This gives us

$$V_{t+1} = V_t [(1 + s)x + (1 - s)(1 - x)] \quad (1)$$

Where V_t is the wealth of the agent at time t and $x \sim \text{Bernoulli}(u)$. Agents seek to maximize the long-term growth rate of their wealth over many time periods T . An agent's expected wealth at the end of its lifetime can be written as

$$V_T(s) = V_0 \prod_{t=1}^T \exp[g_t(s)] = V_0 \exp \left[\sum_{t=1}^T g_t(s) \right] \quad (2)$$

with $g_t(s) = \log(1 + s)$ if successful or $g_t(s) = \log(1 - s)$ otherwise.

Written this way, we can see that the growth rate of V_T depends on a sum of random variables, given by $\log \frac{V_T}{V_0} = \sum_{t=1}^T g_t(s)$. To find the average change in wealth in one time period, we divide this sum by T . As T grows large, by the Law of Large Numbers, this average log growth rate of V_T converges to the long-run log geometric mean of the outcomes, given by

$$\begin{aligned} G(s) &= \lim_{T \rightarrow \infty} \frac{\log \frac{V_T}{V_0}}{T} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T g_t(s) \\ &= u \log(1 + s) + (1 - u) \log(1 - s) \end{aligned} \quad (3)$$

so that the long-run growth can be approximated by

$$V_T(s) \approx V_0 \exp \left[T \cdot G(s) \right] \quad (4)$$

for large enough T . In other words, finding an optimal stake s^* that maximizes the expected long-term growth rate of wealth merely requires maximizing $G(s)$. This leads to an optimal stake that can be calculated explicitly. Popularly known as the Kelly criterion or Kelly bet (Poundstone, 2010) after its initial derivation by Kelly (1956), for our simple model it takes the following form:

$$s^* = \begin{cases} 2u - 1 & : u \geq \frac{1}{2} \\ 0 & : u < \frac{1}{2} \end{cases} \quad (5)$$

This means that a decision-maker who knows the rate of success u can maximize their growth rate by following this formula, and should bet nothing if their *environmental edge* (the extent to which an environment provides an advantage to the gambler, which we denote $\epsilon = u - \frac{1}{2}$) is equal to or less than 0. Indeed, it is easy to show that an equal number of successes and failures yields an overall loss in the agent's wealth: note that the payoff of an agent with n successes and n failures

is $V_0(1+s)^n(1-s)^n = V_0[(1+s)(1-s)]^n = V_0(1-s^2)^n$, which is necessarily less than or equal to V_0 . Losses therefore carry more weight than gains in this dynamic, so it is important to be playing with a favorable edge. This is reflected in the expression for s^* : a success rate of $\frac{1}{2}$ yields an optimal Kelly bet of 0.

Since the above process involves optimizing an exponential growth rate, in an expected utility framework optimal decision-making happens when individuals' psychologies are represented by a logarithmic utility function (Bernoulli, 1954; Levy, 2010, 2024); that is, when the expected utility of a gamble is evaluated as the expectation of the logarithms of its possible outcomes, a notion which aligns with the concept of geometric mean fitness in evolutionary biology (Gillespie, 1974). In the theory of expected utility, an agent's risk preference is given by the curvature of their utility function (i.e., a risk averse agent will perceive the value of gamble outcomes as having diminishing marginal returns). A risk-neutral decision maker (i.e., one using a linear utility function) would choose to bet *all* of their wealth on every gamble with a favorable edge, a strategy that can quickly lead to ruin as soon as the first disfavorable outcome happens. A decision maker with a concave utility function chooses to bet less than all of their wealth, with the chosen stake being determined by the concavity of the utility function. The logarithmic utility function is the one that best fits problems involving multiplicative growth, at least in unconstrained cases with perfect information. A utility that is less concave will lead to over-betting, and a utility that is more concave will result in under-betting. We therefore assume individuals start life *as if* they possessed logarithmic utility functions as a baseline level of risk aversion. This represents an evolutionarily-optimized adaptive risk aversion that always leads to choosing the optimal stake, given perfect information about the environmental edge and given that there are no absorbing boundaries in place. These two assumptions are critical. As we show below, imperfect information about environmental edges and absorbing boundaries add uncertainty that (potentially) requires agents to adjust this initial degree of risk aversion.

Absorbing boundaries: meeting a terrible fate

Risky gambling behavior may involve a string of unlucky decisions that lead to ruin. Wealth can crash, decreasing to a level from which it becomes impossible to recover within a lifetime. This "point of no return" is also known as an *absorbing boundary*, which we denote V_B , such that if wealth ever falls under this value during the T time periods of an agent's lifetime, the agent cannot continue gambling and their wealth is set to zero. We say that such an agent is *ruined*. Absorbing boundaries are a fundamental real-life constraint that all decision-makers must face, whether they be investors navigating a financial landscape or members of a living species adjusting their reproductive patterns in the face of environmental stochasticity.

When $V_B = 0$, the absorbing boundary can only be reached when an agent chooses to stake it all ($s = 1$), such that the first disfavorable outcome will lead to total loss of wealth. $V_B = 0$ is also the scenario for which Equation 5 actually represents the optimal stake. However, individuals often cannot continue taking risky gambles even if they have not technically lost everything, because a baseline amount of resources is required to support basic needs. We represent these scenarios by considering cases where $V_B > 0$.

Throughout the paper, unless stated otherwise, we fix $V_B = 1$. Our minimum starting wealth will thus be $V_0 = 1$. When agents attempt to optimize their resource growth rates under this scenario, a positive absorbing boundary leads to optimal stakes below those indicated by Equation 5. Strictly speaking, the placement of the absorbing barrier is arbitrary; the important thing for agents' wealth is not the absolute location of the barrier, but rather their relative distance from it.

Adding non-zero absorbing boundaries to the model introduces the need for numerical simulation in order to rigorously characterize aspects of the system's behavior, such as the payoff distribution.

However, we can use tools from the theory of martingales in order to find a Brownian approximation of the risk of ruin p_{ruin} , which is given by the following (see Corollary 2.4 in Asmussen and Albrecher (2010), p. 26):

$$p_{\text{ruin}}(s) \approx \left(\frac{V_B}{V_0} \right)^{\frac{2\mu}{\sigma^2}} \quad (6)$$

where

$$\mu = G(s) = u \log(1+s) + (1-u) \log(1-s) \quad (7)$$

and

$$\sigma^2 = u \left[\log(1+s) - \mu \right]^2 + (1-u) \left[\log(1-s) - \mu \right]^2 \quad (8)$$

Equation 6 implies that it is the ratio of V_B to V_0 , and not their individual magnitudes, that has an influence over the probability of ruin. Thus, we define the *wealth buffer* as

$$\aleph = 1 - \frac{V_B}{V_0} \quad (9)$$

which measures the extent to which initial wealth V_0 exceeds the absorbing boundary. This represents a safety net that gives agents a cushion during difficult times (e.g., inherited wealth and/or stable institutions providing a social safety net). When $\aleph = 0$, then $V_0 = V_B$ and the focal agent starts out ruined, while a focal agent with $\aleph = 1$ has an effectively infinite wealth buffer since this implies $V_0 = +\infty$ as long as $V_B > 0$. In this case wealth growth rates are undefined. For this reason, we restrict our use of $\aleph = 1$ to the case of zero absorbing barriers $V_B = 0$. This way, the wealth buffer $\aleph$, ranging from 0 to 1, fully parameterizes the effects of absorbing barriers.

To understand $\aleph$ intuitively, consider the following. Given a starting wealth V_0 , how many losses in an uninterrupted string of losses will lead to ruin? Formally, if we represent the number of losses as B , we get this equation relating V_0 and V_B :

$$V_B = (1-s)^B V_0 \iff \frac{V_B}{V_0} = (1-s)^B \quad (10)$$

The ratio $\frac{V_B}{V_0}$ is equal to the proportion of initial wealth an agent betting s and starting at V_0 has after an uninterrupted string of losses that eventually leads to ruin. This means the wealth buffer can be written as

$$\aleph = 1 - (1-s)^B \quad (11)$$

In other words, for a fixed stake s , the wealth buffer $\aleph$ is the proportion of initial wealth that must be lost in an uninterrupted string of gambles in order to become ruined.

In uncertain environments with absorbing boundary $V_B = 0$ ($\aleph = 1$), the optimal stake is given by Equation 5, but this changes once we allow for non-zero absorbing boundaries. We can also derive the optimal stake (the Kelly bet) for cases involving absorbing boundaries by numerically solving the following expression:

$$s_{\text{ruin}}^* = \arg \max_s \bar{V}(s; \epsilon, \aleph) \quad (12)$$

where

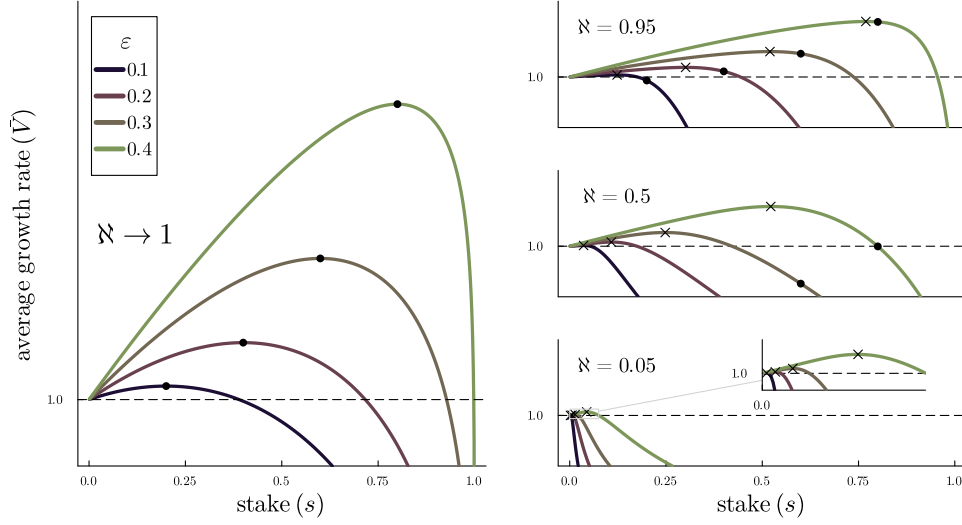


Figure 1: Average geometric wealth growth rates $\bar{V}$, as a function of stake (s). The three plots on the left show the average geometric growth rate of wealth, incorporating the effects of absorbing barriers. The stakes given by Equation 5 are shown as black dots, while the growth-maximizing stakes given by Equation 12 are indicated by an X. Without absorbing boundaries (right plot), growth rate is maximized according to Equation 5.

$$\bar{V}(s; \epsilon, \aleph) = \exp \left([1 - p_{\text{ruin}}(s; \epsilon, \aleph)] \cdot G(s) \right). \quad (13)$$

which converges to Equation 5 in the limit of $\aleph \rightarrow 1$. This is an example of optimization with respect to a “safety-first expected utility” model, which assumes agents make decisions based on both the expected utility of gambles as well as their risk of hitting an absorbing boundary (Levy and Levy, 2009). Figure 1 illustrates the average growth rate as a function of stake for several values of the environmental edge (ϵ) and the wealth buffer ($\aleph$). Optimal stake decreases with both environmental edge and wealth buffer. It is clear that when absorbing boundaries are non-zero and wealth buffers are finite, adopting risk attitudes (stakes) that are optimized for boundary-less scenarios will likely lead to ruin due to overbetting. Wealth buffers matter, as they do in the real world (Smith et al., 2005; Rodems and Pfeffer, 2021; Miller et al., 2021).

Given that the wealth growth rates resulting from risk attitudes depend on both the nature of the environment (the environmental edge) as well as individual wealth buffers, optimizing risk attitudes requires the acquisition of quite a lot of information. Given that the wealth growth rates resulting from risk attitudes depend on both the nature of the environment (the environmental edge) as well as individual wealth buffers, optimizing risk attitudes requires the acquisition of quite a lot of information. Subsequent analyses will consider what happens when agents face non-zero absorbing boundaries, and have no initial knowledge of the environmental edge or the extent of their wealth buffer. Hence, they have no direct knowledge of the risk of ruin or of the optimal stakes they should bet under their particular conditions. Instead, they must try to infer these through learning via direct experience as well as through social transmission. In particular, there are a number of key social cues that an adaptive risk-psychology can plausibly attend to: the success rates in a risk-free

developmental learning period, the rate of ruin among peers, and the beliefs of successful members of the previous generation.

Learning strategies and wealth growth

We assume that agents are not born with pre-encoded information about their environmental edge. Instead, they must infer the edge from individual experience and social learning. When the environmental edge is positive but close to 0, environments are very uncertain: the expected number of successful gambles is nearly equal to the expected number of failed ones. The variance of outcomes is higher, and a “lucky streak” can therefore lead to a biased estimate of the environmental edge. A higher edge means less uncertainty and, consequently, easier learning.

To incorporate learning throughout the lifetime, we assume agents update a belief distribution over the environmental success rates, given by $\Gamma_i = \text{Beta}(a_i, b_i)$, which always starts as a uniform distribution, $\text{Beta}(1, 1)$. The Beta distribution permits iterated learning using straightforward Bayesian updating. As the agents incorporate information from different sources, they update their belief distribution by adding positive values to their Beta’s parameters, where additions to a represent encountering information about successes and additions to b represent encountering information about failures. The precise way that information is incorporated into the posterior updates will depend on the learning strategies, described below.

We rely on numerical simulation throughout the remainder of the paper in order to explore the effects of learning on the stakes that agents choose. When an agent i uses a learning strategy L_i , we summarize their overall success by computing the average growth rate of their wealth achieved with that strategy. This is given by

$$V_T(L_i|\aleph, \epsilon) = \exp \left[\frac{1}{T+1} \sum_{t=0}^T g_t(L_i|\aleph, \epsilon) \right] \quad (14)$$

where the notation indicates that the agent’s growth rate depends on the learning strategy they choose as well as their wealth buffer and the uncertainty in their environment. Because learning is a stochastic process, two agents employing the same learning strategy will not necessarily end up using the same stake after a finite number of gambles. We thus simulate N agents for every learning strategy implemented, and take the average geometric growth rate as an aggregated measure of the learning strategy’s success:

$$\bar{V} = \frac{1}{N} \sum_{i=1}^N \bar{V}_T(L_i|\aleph, \epsilon) \quad (15)$$

An average growth rate of less than 1 indicates that agents using a particular learning strategy are, on average, experiencing losses and wealth shrinkage, while an average growth rate greater than 1 means that on average, agents’ wealth is increasing. Below, we consider how learning throughout the lifespan can lead to the adoption of optimistic or pessimistic risk attitudes.

2.2 The development and cultural transmission of risk attitudes

Here, we establish a full dynamic model for the development and cultural transmission of risk attitudes to consider the long-term effects of different learning strategies on risk attitudes in a population over generational time. This model involves the initial acquisition of risk attitudes during a juvenile period involving risk-free exploration of the environment (*individual learning*), information-sharing between generational peers coupled with socio-environmental feedback from observations of

peer ruin during adulthood (*horizontal social learning*), and information inheritance from previous-cohort elders (*vertical and oblique social learning*). We use this model to also consider the effects of socioeconomic stratification, social learning biases, and dynamic changes in environmental conditions. In the remainder of this section, we describe this model in detail. In the following section, we present the results of our investigative simulations.

Juvenile learning phase: acquiring risk attitudes using intra-generational learning

Consider a population of N individuals in a young generational cohort who all experience an initial wealth buffer $\aleph$, such that they have starting wealth $V_0 = \frac{1}{1-\aleph}$. Perhaps they are the children in a population of migrants who have entered mysterious new lands full of opportunity and, potentially, danger. For this first generation, let us assume that information from elders is unavailable (or so obviously useless that it is ignored). The young cohort must therefore interact with the environment to learn its ways. Subsequent generations will be able to learn from the knowledge acquired by their elders.

In their juvenile phase, agents can sample the environment without actually staking anything. This risk-free sampling is meant to represent individual learning in early development: juveniles get to “play” in risk-free settings and use that information to perform in settings where stakes actually matter. Such low-risk play during childhood is common humans and other complex social animals (Smaldino et al., 2019; Burghardt et al., 2024). The objective is to estimate the environmental edge, ϵ . We divide the *juvenile learning phase* into two key processes: individual learning and peer influence.

1. **Individual learning:** Agents draw τ samples from the environment, representing independent and consequence-free experiences of would-be success or failure. That is, in each of τ time periods during their juvenile stage, agents throw a biased coin, where the probability of success is $u = \epsilon + \frac{1}{2}$. The value of τ represents the length of risk-free juvenile periods. Each agent i adds its observed number of successes w_i during its juvenile period to the a parameters of their belief distribution, and the number of failures $l_i = \tau - w_i$ to the b parameter. This results in a beliefs given by the random variable

$$\Gamma_i^I = \Gamma_i'^0 \sim \text{Beta}(1 + w_i, 1 + l_i)$$

at the end of the juvenile learning period. This distribution is the agent’s initial estimate of the environmental risk, u .

2. **Peer influence:** After individual learning is finished, each agent observes the behavior of n other agents chosen at random from within their cohort (henceforth their *peer learning set*), from which they further update their estimates of u . Such learning from peers is often called “horizontal” transmission in the cultural evolution literature (Cavalli-Sforza and Feldman, 1981). We assume agents incorporate information from their peer learning sets by considering the sum of their peers’ environmental samples:

$$\Gamma_i^P = \Gamma_i'^I \sim \text{Beta}(a_i^P, b_i^P) \tag{16}$$

$$a_i^P = 1 + w_i + \beta_i \sum_{j \in P} w_j \tag{17}$$

$$b_i^P = 1 + l_i + \beta_i \sum_{j \in P} l_j \tag{18}$$

j indexes individuals in i 's peer learning set P , and $\beta_i \in [0, 1]$ is the *weight of peer influence*, which is the extent to which an individual's belief distribution changes after sampling peers at the end of juvenile learning. Peer influence thus works as a form of reinforcement of the information encountered during risk-free individual learning. Because we are interested in the evolution of risk-learning strategies, we allow β_i to evolve in some of our analyses.

After the juvenile learning phase, agents apply their risk attitudes to gambles that have consequences, during which time they can continue to update their risk attitudes. To simplify notation, we will refer to the beliefs about environmental risk at gambling period t as the random variable $\Gamma_{i,t}$ and the parameters of the (Beta) belief distribution it follows as $a_{i,t}$ and $b_{i,t}$. This means that in the absence of information from elder cohorts, the beliefs at the end of the juvenile period, Γ_i^P , are initially used to face the perils of adulthood. Symbolically, $\Gamma_{i,0} = \Gamma_i^P$.

Gambling phase: adjusting risk attitudes in response to observed ruin

At the end of their risk-free exploratory juvenile period, individuals venture into the world and use their beliefs about risk to make meaningful gambles. They play one gamble for each of T periods, all while keeping an eye on their peer learning set. Gambles are played by sampling the beliefs $\Gamma_{i,t}$ every time period t , obtaining an estimate of u (which we call $\hat{u}_{i,t}$), and passing this estimate through Equation 5 to determine their stake. While wealth, as a social phenomenon, is present in subsistence economies (Mulder et al., 2009) and even in non-human systems (Strauss and Shizuka, 2022), what can be considered an absorbing boundary or point of "ruin" is a more elusive concept that can vary significantly from context to context, whether that mean different societies at a single point in time or the same society across generations (for example, one need not arrive a point of starvation before any cultural influence on the next generation is completely lost). Thus, as stated before, learners have little-to-no knowledge on how wealth buffered they are until they can actually experience ruin in one way or another, be it through direct life experience, through observation of peers, or through information inherited from previous generations (assuming that buffers remain stable). We first assume naive learners could learn only from the environment and peers. We then add the possibility of culturally inherited risk attitudes that can become attuned to wealth buffers. We denote the stake used in period t as

$$\hat{s}_{i,t} = s^*(\hat{u}_{i,t}) \quad (19)$$

Hence, the observable risk attitudes of individuals with less concentrated belief distributions are assumed to correspond to greater variability in their chosen stakes from one time period to the next, product of the higher uncertainty these individuals have about the true value of the environmental edge.

Once the stake has been wagered, each agent performs an independent Bernoulli trial U_t with success probability u and their wealth is updated according to their success or failure. In the log-wealth scale, this is expressed as the following stochastic recursion:

$$g_{i,t+1} = \begin{cases} g_{i,t} + \log(1 + \hat{s}_i) & \text{when } U_t = 1 \\ g_{i,t} + \log(1 - \hat{s}_i) & \text{when } U_t = 0 \end{cases} \quad (20)$$

We allow agents to keep learning from their trials, such that they continue to update their beliefs $\Gamma_{i,t}$ in the same way as they did during juvenile individual learning, iteratively adding successes and failures throughout their lifetime of T periods, so that

$$\Gamma_{i,t+1} \sim \text{Beta}\left(a_{i,t} + U_t, b_{i,t} + (1 - U_t)\right) \quad (21)$$

If at any point in the T periods of the gambling process an agent's wealth falls to or below the absorbing boundary, the agent becomes ruined, their growth rate is set to 0 (g_i is set to $-\infty$), and they make no further gambles. Overly optimistic betting (i.e., overly large stakes) will often lead to ruin. In this case, continued peer learning can help agents avoid their peers' mistakes. In particular, if an agent observes someone go to ruin who had been using a gambling strategy similar to their own, they may adjust their risk attitudes to be less optimistic.

Sensitivity to peer ruin: fear is the risk-killer

At the end of childhood, individuals in this model are likely to have accurately estimated the environmental edge through a combination of individual and social learning. However, the childhood phase does not provide any information about the full risk of ruin (determined not just by the environmental edge, but also by an individual's wealth buffer), and agents following the Kelly criterion implicitly assume that there is no such risk. If their adult lives are heavily buffered by large wealth reserves, agents applying heuristic rules like the Kelly criterion will produce nearly optimal betting behavior. Conversely, if wealth buffers are small, the Kelly heuristic leads to overbetting even with extremely precise estimates of environmental risk. This is fundamentally because optimal behavior implied by Equation 12 requires knowledge of both the environmental uncertainty and of the agent's own wealth buffer, and childhood exploration can only update an agent's estimate of the former. Essentially, a risk-free developmental period has an effectively infinite wealth buffer ($\aleph \rightarrow 1$), so agents will come out of juvenile learning periods with overly-optimistic beliefs. As adults, though, learners can observe ruin events among peers, and so plausibly benefit by incorporating information about how close they are to an absorbing boundary.

The fact that juvenile learning can easily lead to catastrophic overconfidence calls for a mechanism that will aid in pessimistic adjustment throughout the lifetime. Here we add to the model the possibility for peer learning in adulthood, such that agent i reduces their stakes when they observe peers go to ruin as a function of how similar their belief are to i 's beliefs (which they take as a cue that their own risk attitudes are overly optimistic). We assign agents a *sensitivity* $\delta \in [0, 1]$, which is the extent of their response to observing ruin among peers. We remind the reader that $a_{i,t}$ and $b_{i,t}$ represent the accumulated positive and negative beliefs (respectively) of agent i due to all information influences up to time period t . Let k index the peer ruin events i observed in time period t after gambling, and $b_{i,t,k}$ are the negative beliefs after observing k peer ruin events in time period t , such that $b_{i,t} = b_{i,t,0}$ are the negative beliefs after gambling but before observing any peer ruin event. Pessimistic adjustment thus behaves according to the following recursion:

$$b_{i,t,k+1} = \left[1 + \zeta \left(\frac{|\bar{s}_j - \bar{s}_i|}{\max[\bar{s}_i, 1 - \bar{s}_i]}, \delta_i \right) \right] \cdot b_{i,t,k} \quad (22)$$

where $\bar{s} = s^*(\frac{a_t}{a_t + b_{t,k}})$ is an agent's mean stake after observing k ruin events at time t , i indexes the focal agent, j indexes the observed ruined peer (whose mean stake is "frozen" in place after they hit ruin, for convenience), and ζ is the *social trauma function* given by

$$\zeta(x, \delta) = \begin{cases} (1 - x)^{\frac{1}{2\delta}}, & \text{if } 0 < \delta < \frac{1}{2} \\ 1 - x^{\frac{1}{2(1-\delta)}}, & \text{if } \frac{1}{2} \leq \delta < 1 \end{cases} \quad (23)$$

This function is meant to capture the idea that agents will react to the observed ruin of their peers by becoming more pessimistic, and that they will be affected more heavily by peers with more similar risk beliefs. Their sensitivity δ controls the strength of their pessimistic response as a function

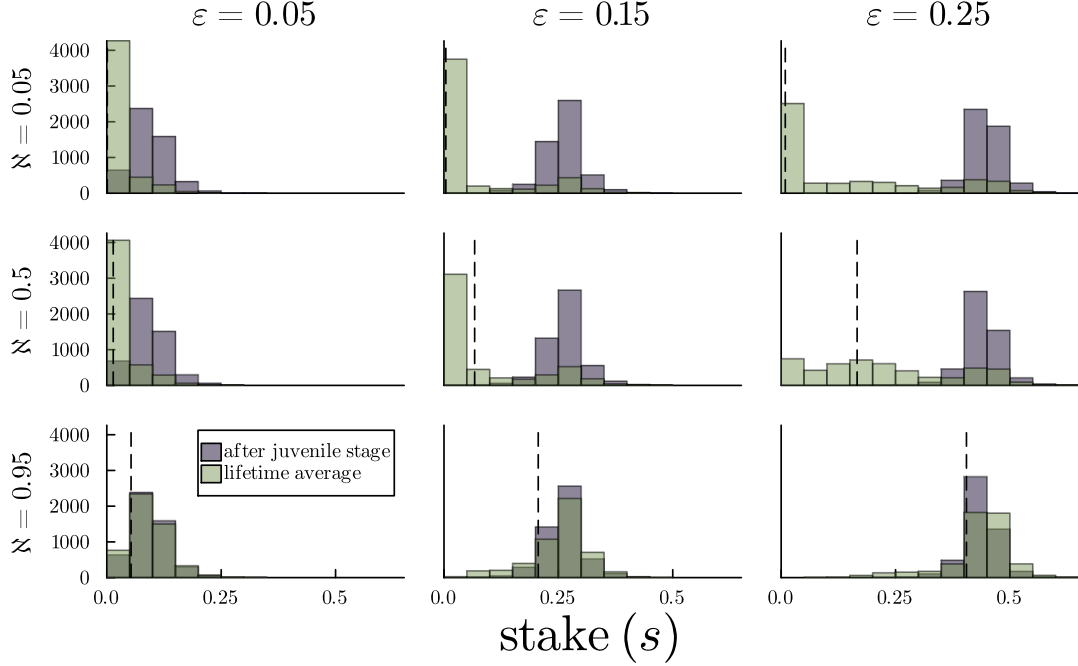


Figure 2: Distributions of expected stakes ($\bar{s}$) at the end of juvenile learning stages (gray) and at the end of gambling (green) when agents have a social trauma response given by their sensitivity to peer ruin. High risk of ruin scenarios (low edge, low wealth buffer) lead to a higher divergence between the distributions, indicating higher optimism at the end of juvenile learning and, as a consequence, a greater degree of acquired pessimism during the lifetime. $N = 10^4$ agents were simulated going through the juvenile learning phase with $\tau = 15$ and $\beta = 0.15$. They then experience $T = 100$ rounds of gambling while sampling their belief distribution and adjusting their beliefs according to a sensitivity of $\delta = 0.1$. The optimal stake given by Equation 12 is plotted as a vertical dashed line for each case.

of how similar in beliefs observed ruined peers are. As $\delta_i \rightarrow 1$, fully sensitive agents double their present failure count when they observe a ruin event, unless the observed ruined peer is maximally different to them in beliefs, in which case they do not react at all. When $\delta \rightarrow 0$, the focal agent is only affected by the observed ruined of a peer whose mean stake is exactly equal to theirs, ignoring all other observed peer ruin events. $\delta \in (0, 1)$ interpolates the cases between these two extremes, allowing for a full continuum of sensitivity to observed ruin, allowing for different strengths of social trauma response. In Figure S2 we plot this function, showing its symmetry on the domain of δ and its convergence to Dirac delta functions at the limits of $\delta \rightarrow 0$ and $\delta \rightarrow 1$, properties that make it a suitable tool for measuring sensitivity to observed peer ruin.

Figure 2 plots the mean stakes of agents simulated coming out of juvenile learning and at the end of their lifetimes for a fixed value of sensitivity. It shows how, as wealth buffers and environmental edges decrease, agents start their lives with more overly-optimistic risk attitudes, and thus require higher degrees of pessimistic adjustment to get closer to the optimal stake. It also shows how high adversity can lead to over-adjustment, generating extreme levels of pessimism that aid in safety but mitigate any future growth (see i.e. middle row, with $N = 0.5$).

Cultural inheritance: using information from those who came before

We have so far ignored the value of information from previous generations, and we have therefore ignored the possibility of culturally-inherited risk attitudes. Yet there are good reasons to expect such capacities to be adaptive in this model. Elders from the previous generation have already experienced the adult environment, and have updated their risk preferences accordingly. Since offspring are likely to face the same environmental uncertainty with the same wealth buffers as their parents, attending to elder risk preferences at the end of childhood could plausibly yield adaptive benefits. This is especially true for estimating one’s wealth buffers, which is impossible in childhood. Let us assume now that agents do have access to this source of information, showing how this idea can be modeled and added to our current setup.

Imagine that our first generation of agents, which relied solely on individual and peer learning, has given way to a next generation. The old generation retires from productive activity, but they still stay around to give advice to their new generation. For simplicity, we assume individuals in this next generation inherit their parents’ initial wealth buffer and undergo the same juvenile learning period described above. Here, however, we add the possibility of a new learning stage, *elder influence*, which in our model implementation happens after juvenile learning but before the gambling stage. However, we note that the order of these stages does not matter for the resulting adulthood belief distribution, since each stage merely adds values to the belief distribution parameters and thus can be permuted freely. This adds a dimension of age structure to our model, which has been shown to influence dynamics of cultural transmission (Kandler et al., 2023).

We implement a simple blending inheritance process in which the focal agent observes m randomly selected agents from the previous generation (henceforth the *elder learning set* H_i) and takes the average of their accumulated environmental impressions. This includes all the information that the elders in the elder learning set obtained from individual and social learning throughout their lifetimes, and thus includes their estimate of the central tendency of the environmental edge as well as the uncertainty they might have about it. Note that, if $m = 1$, this is equivalent to the sort of unbiased learning rule used in many cultural evolutionary models. While peer influence is about the social reinforcement of within-cohort experiences, elder influence, as an inheritance process, can be the main source of information individuals use when starting their actual interactions with the world. In our model, this means that elder information can “overwrite” the information from juvenile experiences, to a greater or lesser degree. And if the received elder information is well-attuned to the receiver’s conditions, then it can be of a higher quality than information sampled through a noisy and overshoot-prone process of risk-free juvenile learning, potentially saving them from early ruin due to over-optimism.

We assume that agents ignore elders who have gone to ruin, as their information may be deemed low-quality—this can be viewed as a sort of cultural viability selection. Also referred to as “demographic filtering”, this selective process directs learning towards older age classes that can serve as reservoirs of adaptive behavior (Deffner and McElreath, 2022). The elder learning set H_i is thus formed by sampling m elders from the population and “dropping” those with an average growth rate of 0. If, by chance, an agent ends up sampling only ruined agents, then no elder learning set is formed and no elder influence occurs.

At the end of their juvenile period, agents begin adulthood with estimates of the environmental edge drawn from two sources: intra-generational learning (individual and peer learning), and inter-generational (elder) learning. From generation 2 onwards, agents at the end of their juvenile phase begin adulthood with beliefs $\Gamma_{i,0}$ that weight these two influences:

$$\Gamma_{i,0} = \Gamma_i^{\prime P} \sim \text{Beta}(a_{i,0}, b_{i,0}) \quad (24)$$

where

$$a_{i,0} = a_i^P + \alpha_i \frac{1}{|H_i|} \sum_{k \in H_i} a_{j,T} \quad (25)$$

and

$$b_{i,0} = b_i^P + \alpha_i \frac{1}{|H_i|} \sum_{k \in H_i} b_{j,T} \quad (26)$$

where k indexes individuals in i 's elder learning set H_i , $|H_i|$ is its cardinality (number of elements in the set), and $\alpha \in [0, 1]$ is the *weight of elder influence*, which determines how much agents rely on intra- versus inter-generational learning. We refer to learning strategies that heavily rely on elder influence (α close to 1) as *conservative*, as this conserves the knowledge accumulated by previous generations, and a high reliance on inter-generational learning is characteristic of cultural conservatism (Morin, 2022).

The dangerous allure of payoff bias

While we focus on a simple form of demographic filtering in elder influence, which simply ignores all individuals who have experienced ruin, another form can be implemented by introducing an explicit payoff bias to cultural learning. For a payoff-biased social learner, the higher the wealth of an observed elder, the higher the probability of adopting their risk attitude. While payoff-biased social learning can outperform unbiased learning if payoff is reliably associated with the adoption of adaptive behaviors (McElreath et al., 2013), payoff-biased learning can also be problematic for learning risk attitudes in uncertain environments. This is because those agents with the highest payoffs will tend to be those that made large gambles and got lucky, even though most individuals gambling with similarly optimistic stakes will experience ruin. By focusing on these rare successes, agents using payoff-biased social learning may adopt excessively optimistic risk attitudes, exposing themselves to significant risks of ruin (Van Tilburg and Mahadevan, 2020; Baldini, 2012). On the other hand, in a socioeconomically stratified society, success can act as a *de facto* social marker for the wealthy classes, directing social learning to targets of similar wealth (Smaldino and Pérez Velilla, 2024).

For some analyses, we implemented payoff bias in elder influence by assuming that learners selectively learn from the agent in their elder learning set with the highest average growth rate, ignoring the rest. The elder learning set H_i therefore only contains the highest-growth rate individual. Belief update then proceeds according to Equation 24. Symbolically, we denote by $\mathcal{L}_i \in \{\text{UB}, \text{PB}\}$ the *learning bias* of agent i , with UB being shorthand for *unbiased blending* and PB for *payoff bias*.

Group structure and parochialism

So far we have assumed that populations face homogeneous environments. However, many real-life populations, especially those in contemporary settings, exhibit class stratification and other forms of structural inequalities. A working class individual often faces a more adverse risk landscape than accommodated bourgeoisie. If we assume that agents only learn from their fellow class members, then we can simply model stratification as multiple instantiations of the base model described above. However, in order to allow the possibility that agents learn from others outside of one's own class but exhibit some preference for their own class, we add another behavior to the learning strategy: parochial (or similarity-biased) social learning (Smaldino and Pérez Velilla, 2024).

We model class stratification by creating two classes of agents, each of which experiences a class-specific environmental edge and initial wealth buffer. Agents are assigned an observable class tag $C_i \in \{c_0, c_1\}$, which are represented in the population with frequencies f and $1 - f$, respectively. We use c_0 to refer to the advantaged group (henceforth *bourgeoisie*) and c_1 to the disadvantaged group (*working class*). Parochialism is then implemented by a binary variable $\mathcal{P}_i \in \{0, 1\}$, so that if $\mathcal{P}_i = 1$, an agent is parochial and drops any sampled outgroup members from their learning sets. Note that this implies an endogenous cost to parochialism, as parochial agents can end up with fewer agents in their learning sets than non-parochial counterparts.

Cultural evolution and its consequences

After learning and cultural inheritance, agents proceed with the gambling phase, just as their predecessors did. At the end of their gambling, and thus their lifetime, we collect their average growth rates and use them to generate the next cohort of individuals. The model simulates cultural evolution through a fitness-based evolutionary algorithm, where individuals adapt learning strategies over generations. Each individual adopts a learning strategy represented by a vector $\mathbf{L}_i = \{L_{i,1}, L_{i,2}, \dots, L_{i,k}\}$, where $L_{i,j}$ corresponds to specific components of their strategy (such as their weights for peer and elder influence, sensitivity, parochialism and payoff bias). Fitness is calculated by normalizing each agent’s average growth rate at the end of their lifetime by the sum of all growth rates in the population (or in their respective class, if the population is stratified), using a machine epsilon lower bound for numerical stability.

Strategy inheritance occurs stochastically. A new generation of size N is spawned and the components of their learning strategy vector $\mathbf{L}_i$ are randomly sampled from previous-generation individuals. The probability that a component of the learning strategy vector is inherited from a particular previous-generation individual is given by said individual’s cultural fitness, with mutation introducing variation. Mutations may involve random perturbations for continuous strategies or replacement for discrete strategies. Continuous strategies evolve through Gaussian mutations bounded to feasible ranges, while discrete strategies are sampled from predefined sets. The model incorporates parameters such as mutation rates (μ) and noise levels (σ), ensuring controlled exploration of the strategy space. Only the most recent generation goes on to gamble, while the previous one retires to enjoy the fruits of their labor.

Simulations run over 2,500 generations to capture long-term dynamics, and in cases where there is population class stratification, we assume individuals inherit learning strategies within their class only. This approach models cultural adaptation as a balance between exploiting successful strategies and exploring novel ones through mutation, reflecting the mechanisms of cultural transmission under varied environmental and social conditions. We summarize the steps for how agents do their learning in Model Box 1. Having done so, we allow the learning strategies of agents to evolve, following the evolutionary algorithm informally described above and outlined in Model Box 2. We present the results of the evolution of learning strategies in a progressive manner, starting with only within-generation learning, then adding the possibility of elder influence and payoff bias, and finally allowing for group structure and parochialism. In this way, it becomes clear how the addition of age structure and class stratification affect the evolution of learning strategies and, in consequence, risk attitudes.

3 Analyses and results

For all results, 25 randomly-seeded simulation runs were performed per parameter combination. The key parameter we vary in all analyses is the wealth buffer $\aleph$, performing simulations for values between 0.05 and 0.95 in steps of 0.05. For discrete social learning strategies (learning bias,

Model Box 1: Initialization and Learning Dynamics

Parameters:

- N : Number of agents
- V_0 : Initial wealth; determined by $\aleph$
- V_B : Absorbing boundary (default: 1)
- $\aleph \in (0, 1)$: Wealth buffer, $\aleph = 1 - V_B/V_0$
- $u \in (0.5, 1)$: Environmental success probability
- ϵ : Environmental edge ($\epsilon = u - 0.5$)
- T : Time periods in the gambling phase
- τ : Number of risk-free samples during the juvenile phase
- n : Number of sampled peers
- m : Number of sampled elders (generation 2+ only)
- $f \in (0, 1)$: relative frequency of class g_0 individuals in a stratified population.

Agent Properties:

- $\log_i$: Logarithm of wealth for agent i at time t
- $\Gamma_{i,t}$: Random variable obeying a belief distribution over u at time t ($\text{Beta}(a_{i,t}, b_{i,t})$)
- $w_{i,t}, l_{i,t}$: Success and failure counts during learning
- P_i : Peer learning set for agent i
- H_i : Elder learning set for agent i (generation 2+ only)
- $\hat{u}_{i,t} \in [0, 1]$: Estimated success probability for agent i at time t
- $\hat{s}_{i,t} \in [0, 1]$: Stake for agent i at time t , based on $\hat{u}_{i,t}$

Agent Learning Strategies:

- $\alpha_i \in [0, 1]$: Weight of elder influence of i
- $\beta_i \in [0, 1]$: Peer influence weight of i
- $\delta_i \in [0, 1]$: Sensitivity to peer ruin of i
- $\mathcal{L}_i \in \{\text{UB}, \text{PB}\}$: Learning bias of i
- $C_i \in \{c_0, c_1\}$: Class membership of i
- $\mathcal{P}_i \in \{0, 1\}$: Parochialism of i

Algorithm:

1. Initialization:

- If there is class structure, give each individual a class tag $C_i \in \{c_0, c_1\}$ with probability f and $1 - f$ respectively.
- Set initial wealth: $g_{i,0} = \log(V_0)$ for all agents, where $V_0 = 1/(1 - \aleph)$. If there is class structure, V_0 is calculated with the class-specific $\aleph$ value.
- Initialize beliefs: $\Gamma_i^0 = \text{Beta}(1, 1)$ (uniform prior).
- If first generation, agents are given random values as the strategies in their learning strategies.

2. Juvenile Learning:

- Draw τ_i outcomes from $\text{Bernoulli}(u)$. If there is class structure, use the class-specific u value.
- Update Γ_i with successes (w_i) and failures (l_i).
- Peer influence: Choose n peers to form P_i , and add their successes/failures weighted by β_i (Equation 16). If parochial, drop any sampled out-group members.

3. Elder Influence (generation 2+ only):

- Observe m previous generation elders at random to form H_i , dropping any sampled ruined individuals. If parochial, drop any sampled outgroup members. Integrate beliefs using elder influence weight α_i (Equation 24).

4. Gambling Phase: For $t = 1$ to T :

- Sample $\hat{u}_{i,t}$ from $\Gamma_{i,t}$.
- Compute optimal stake: $\hat{s}_{i,t} = 2\hat{u}_{i,t} - 1$ if $\hat{u}_{i,t} \geq 0.5$, else $\hat{s}_{i,t} = 0$.
- Sample a $U_t \in [0, 1]$ from $\text{Bernoulli}(u)$ and update wealth:

$$g_{i,t+1} = \begin{cases} g_{i,t} + U_t \log(1 + \hat{s}_{i,t}) \\ g_{i,t} + (1 - U_t) \log(1 - \hat{s}_{i,t}) \end{cases}$$

- Update belief distribution using successes and failures (Equation 21).
- Peer ruin: update beliefs using $\zeta(|\bar{s}_i - \bar{s}_j|, \delta_i)$ to penalize $b_{i,t}$ if peer ruin occurs (Equation 22).

5. Cultural Selection and Iteration:

- Compute final wealth: $V_i = \exp((g_{i,T} - g_{i,0})/T)$ if not ruined, else $V_i = 0$.
- Selection, spawning new generation and mutation (see Model Box 2).

Model Box 2: Evolutionary Algorithm with Learning Strategies

Explanation:

- The evolutionary algorithm operates on a population of N individuals, each with a learning strategy $\mathbf{L}_i = \{L_{i,1}, L_{i,2}, \dots, L_{i,k}\}$, where $L_{i,j}$ represents a learning strategy.
- Strategies evolve iteratively through fitness-weighted inheritance and mutation.
- Strategies with higher fitness (better payoffs) are more likely to propagate.
- Mutation introduces diversity by random perturbations or replacement.

Summary of Symbols:

- $\mathcal{F}_i$: Fitness of individual a_i ,
- $L_{i,j}$: Learning strategy j of individual a_i ,
- μ_j : Mutation rate for strategy j ,
- σ_j : Mutation noise (Gaussian perturbation) for strategy j ,
- $\mathbf{L}_j$: Vector of all j -th strategies across the population,
- W_i : Normalized fitness weight of individual a_i ,
- $U(0,1)$: Uniform random distribution over $[0,1]$,
- $\mathcal{N}(0, \sigma_j)$: Normal distribution with mean 0 and standard deviation σ_j .

Default Evolutionary Parameters:

- We run evolutionary simulations for 2500 generations.
- For continuous strategies $\in [0,1]$, we use $\mu_j = 0.001$ and $\sigma_j = 0.01$.
- For discrete strategies with 2 possible values, we use $\mu_j = 0.01$.

Algorithm:

If there is class stratification, the following algorithm describes within-class cultural evolution (we discard migration between classes).

1. Fitness Calculation:

- Compute individual fitness:

$$\mathcal{F}_i = \max \left[\exp \left(\frac{g_{i,T} - g_{i,0}}{T} \right), l \right]$$

with l a machine epsilon to ensure numerical stability.

- Normalize fitness weights:

$$W_i = \frac{\mathcal{F}_i}{\sum_{i=1}^N \mathcal{F}_i}.$$

2. Learning Strategy Vectors:

- Extract vectors for each strategy:

$$\mathbf{L}_j = \{L_{1,j}, L_{2,j}, \dots, L_{N,j}\}.$$

3. Strategy Inheritance and Mutation:

- Generate N new generation individuals.
- Inherit strategy $L'_{i,j}$ with probability $1 - \mu_j$:

$$L'_{i,j} = \text{sample}(\mathbf{L}_j, \mathbf{W}).$$

- Mutate strategy $L'_{i,j}$ with probability μ_j :
 - Uniform mutation:

$$L'_{i,j} = U(0,1).$$

- Gaussian mutation:

$$L'_{i,j} = \text{clamp}(L'_{i,j} + \mathcal{N}(0, \sigma_j), 0, 1).$$

4. Special Cases for Discrete Strategies:

- For predefined strategy sets $\mathcal{S}$:

$$L'_i = \begin{cases} \text{sample}(\mathcal{S}, \mathbf{W}) & \text{if no mutation,} \\ \text{rand}(\mathcal{S}) & \text{if mutation.} \end{cases}$$

5. Iteration:

- New generation learns and plays the game (see Model Box 1).

parochialism), population frequencies are reported. Median values for the average lifetime stake and continuous learning strategies are reported, along with their interquartile ranges.

3.1 Peer influence and sensitivity are favored in the absence of elder information, but lead to ill-fitting risk attitudes

In our baseline scenario, we consider a population of agents that rely solely on juvenile exploration and intra-generational peer interactions to learn about their risky environment, ignoring inter-generational transmission. Specifically, we allow the weight of peer influence (β) and the sensitivity to peer ruin (δ) to evolve, while excluding elder influence ($\alpha = 0$), payoff bias ($\mathcal{L} = \text{UB}$), class stratification and parochialism ($\mathcal{P} = 0$). We also fix $n = 10$ (number of sampled peers) and $\tau = 15$ (individual learning period length) for the plots, but our simulated data shows that the resulting patterns are generally robust for different n and τ values (see code supplement and published data).

Figure 3 shows the resulting evolutionary outcomes for the evolving parameters as a function of the wealth buffer ($\aleph$) under different environmental edges (ϵ). In general, agents benefit from stronger reliance on peer influence. Incorporating information from multiple peers allows them to more accurately estimate the environmental success probabilities even if it could lead to overly optimistic stakes. As wealth buffers increase the value of heavy peer influence declines only slightly, unless the environmental edges are high (i.e. $\epsilon = 0.25$), in which case agents will rely more on individual learning due to the low uncertainty in their environments. This pattern suggests that in particularly harsh and informationally-constrained environments, individual and peer learning, even when imperfect, considerably helps to reduce uncertainty.

The sensitivity to peer ruin (δ) peaks at low wealth buffers, and decreases as wealth buffers increase. When both wealth buffers and environmental edges are high enough (i.e. $\epsilon = 0.25$) this decrease is pronounced, leading to low sensitivity values as wealth buffers increase. Otherwise, sensitivity decreases to intermediate values (~ 0.5). This indicates an evolved balance: ignoring peer ruin when deprived would expose agents to repeated catastrophic failures, while excessive reactivity to peer ruin, when well-buffered and especially under good environmental conditions would stifle risk-taking more than what is necessary. Maintaining a moderate sensitivity thus ensures a cautious yet flexible approach to risk calibration, especially under the over-optimism promoted by high peer influence, but when under very favorable edges and considerable buffering, it is best to become insensitive to the observed ruin of other individuals.

Predictably, the resulting stakes (s) exhibit a strong dependence on the wealth buffer. Under low buffers, agents adopt nearly zero stakes on average across their lifetime, reflecting strong caution. As buffers increase, stakes rise substantially, especially under more favorable edges, due to the decrease in sensitivity with increasing buffers. This pattern confirms that as the environment becomes more predictable and forgiving, agents confidently adjust towards higher levels of risk-taking, approaching (but not necessarily matching) the unconstrained Kelly bet. However, the evolved stakes are not a very good fit to the optimal stake, resulting in heavy over-betting when environmental edges are low and heavy under-betting as they increase. This indicates that agents' learning procedure, even at equilibrium values, is lacking. Peer information is not enough.

In summary, when agents have no access to inter-generational learning or payoff biases, they evolve moderate-to-high levels of peer influence and sensitivity. These conditions lead to cautious initial strategies that become bolder only as wealth buffers and environmental favorability increase, showing how reliance on peer information and moderate responsiveness to failure together foster adaptive, albeit imperfect, risk attitudes under varying environmental conditions and lack of access to other sources of information. However, it is clear that these mechanisms on their own do not foster very accurate risk attitudes. Below, we show that adding elder influence as a strategy resolves

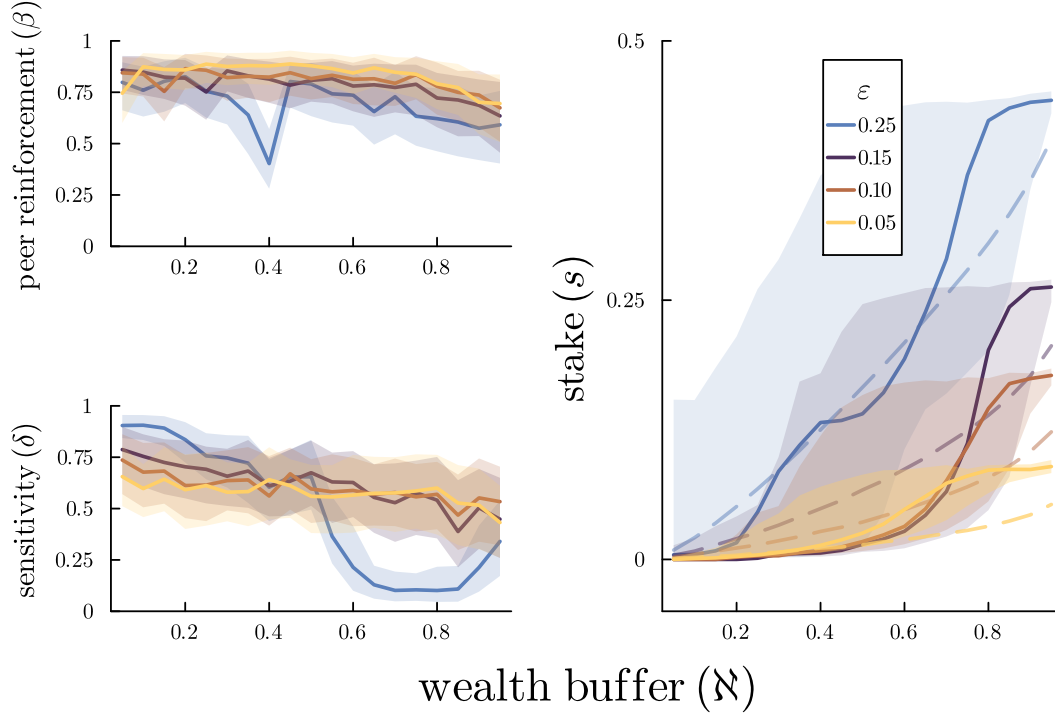


Figure 3: **Results for the baseline scenario.** Agents rely solely on juvenile exploration and peer influence (β), and sensitivity to peer ruin (δ) evolves. Each line corresponds to a different environmental edge (ϵ), with lighter shades representing lower edges. On the left, the top panel shows that peer influence remains relatively high but declines slightly as wealth buffers ($\aleph$) increase. The bottom panel shows that sensitivity to ruin stays moderate and stable, around 0.5–0.7, across conditions, except when environmental edges and wealth buffers are high, in which case sensitivity declines sharply. On the right, the final evolved stakes (s) rise sharply with wealth buffer, indicating that as conditions become more forgiving, agents confidently increase their risk-taking, leading to overly-optimistic stakes at high wealth buffers. The only exception is when environmental edges are very low ($\epsilon = 0.05$), in which case agents evolve highly-optimistic stakes across the range of wealth buffer. Shaded regions represent the interquartile range across simulation runs.

this issue.

3.2 Conservative elder influence undermines exploratory learning as wealth buffers decrease and enables more accurate beliefs about risk

In this scenario, we introduce the possibility of vertical and oblique cultural transmission from older generations. Agents can now incorporate information from elder individuals, allowing the weight of elder influence (α) and the presence of payoff bias ($\mathcal{L}$) to evolve, in addition to peer influence (β) and sensitivity (δ). We fix $n = 10$, $\tau = 15$ and $m = 10$ for these plotted results, but our simulated data shows that the resulting patterns are robust to different parameter values. By enabling elder influence, we investigate how cultural memory and selective attention to successful elders shape risk attitudes across generations. Learning strategies are thus given by $\mathbf{L}_i = (\alpha_i, \beta_i, \delta_i, \mathcal{L}_i)$.

Figure 4 shows the results of these simulations. Compared to the baseline scenario, the introduction of elder influence leads to systematically different evolutionary outcomes in the learning strategies: in general, individual learning and peer influence have a more limited contribution to belief distributions, especially at low wealth buffers, while elder influence becomes a major source of adaptive information, allowing for more accurate risk attitudes.

As wealth buffers ($\aleph$) increase, elder influence (α) generally remains high across the range of wealth buffer, only declining significantly when buffers are very high. When conditions are harsh-to-moderate, relying on the accumulated knowledge of previous generations allows populations to avoid repeated missteps, stabilizing risk attitudes and preventing ruin. Under high wealth buffers, the value of heavy reliance on elder information diminishes, as younger generations can more readily discover successful strategies on their own.

Interestingly, peer influence (β) now shows a more complex pattern, reflecting an interplay between the direct cues gleaned from the environment, social cues from peers, and the inherited information from older cohorts. At lower wealth buffers (< 0.5) the adoption of peer influence depends on the environmental edge: harsher, more uncertain environments promote the use of peer influence, while less uncertain environments select against it. This suggests that the presence of inter-generational knowledge changes the relative importance of peer learning over the range of environmental safety. Only when environments offer very poor information and higher risks does it become advantageous to use the additional information from peers to complement that of elders. Additionally, because elder influence remains high, the overall contribution of peer influence on agents' risk attitudes is also much more subtle.

The previous pattern for sensitivity (δ) remains robust, with high sensitivity at lower wealth buffers which decreases to moderate levels ($\delta \approx 0.5$, implying linear sensitivity) as wealth buffers increase, unless environmental edges are very high (i.e $\epsilon = 0.25$), in which case sensitivity collapses to low values as buffers increase. The stakes (s) again rise with increasing wealth buffers, but when cultural transmission is present, the population can adopt more adaptive stakes (closer to the Kelly optimum) earlier, due to useful initial knowledge from elders.

3.2.1 Payoff bias is a luxury of the rich

When the environmental edge is low and wealth buffers are limited, populations are not helped by payoff-bias social learning, because it often means emulating individuals who were merely lucky rather than genuinely well-adapted. In these high-risk environments, blindly imitating winners frequently leads to ruin. By contrast, when wealth buffers are large and safety nets robust, the cost of being wrong decreases. Under these more forgiving conditions, payoff bias becomes viable, although not necessarily adaptive, allowing populations to amplify success stories, even when doing

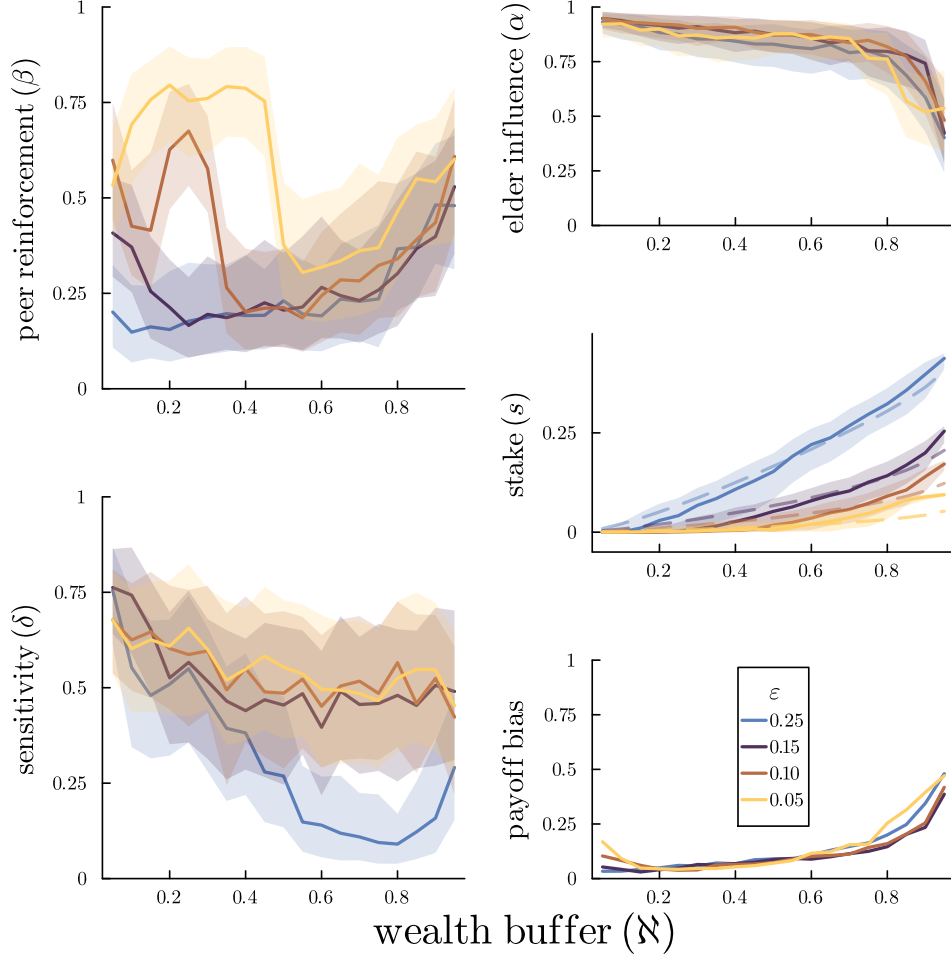


Figure 4: **Results when elder influence is enabled.** Allowing vertical and oblique transmission from previous generations, we now let elder influence (α) and payoff bias ($\mathcal{L}$) evolve alongside peer influence (β) and sensitivity (δ). Each line represents a different environmental edge (ϵ), with lighter shades indicating lower edges. Top-left: Peer influence (β) shows a complex relationship with wealth buffer ($\aleph$), being selected for under high risk scenarios, and selected against as risk decreases. Bottom-left: Sensitivity (δ) remains moderate, indicating balanced responsiveness to peer ruin, decreasing as wealth buffers increase. Top-right: Elder influence (α) starts high in risky conditions but declines as wealth buffers grow, reflecting reduced reliance on ancestral guidance when conditions are safer. Middle-right: Stakes (s) rise with increasing $\aleph$, now reaching values that fit the optimal Kelly stake (dashed lines) better than in the baseline scenario due to better-informed starting conditions. Bottom-right: Payoff bias ($\mathcal{L}$) remains low in dangerous settings but increases somewhat in safer environments, where imitating successful outliers can enhance returns. It increases slightly in particularly impoverished scenarios ($\aleph = 0.05$), due to the disproportionate evolutionary contribution of high-stakes lucky individuals. Shaded regions represent the interquartile range across simulation runs.

so carries a higher risk of acquiring overly-optimistic beliefs. In other words, strong wealth buffers transform payoff bias from a reckless (and likely catastrophic) strategy into a lower-stakes gamble, since agents need not deal with high risks of ruin from the get-go. However, payoff bias is always a high-risk choice relative to unbiased blending, and thus it is never adopted by a majority of the population in homogenous societies.

While payoff bias evolves to remain relatively low under risky conditions, it increases slightly under the deprivation of very low wealth buffers ($\aleph = 0.05$). This is due to the fact that in these extremely impoverished conditions, some lucky individuals who adopted payoff bias and proceeded to use overly-optimistic high stakes did not experience ruin and thus achieved significantly higher growth rates than the median surviving individual, who played it safe. These reckless individuals in turn have a disproportionate, albeit still limited, cultural evolutionary contribution to the learning strategies of the next generation. While most of the contribution to the next generation’s learning strategies will still come from individuals that played it safe and used non-payoff biased learning, the high payoffs of the very few surviving reckless individuals allow them to leave a cultural imprint on the following cohort. In these scenarios, most of the population will exhibit high risk avoidance, but there is a stable fat tail of payoff-biased, highly risk-tolerant agents (see Figure 5). In other words, in deprived populations, individuals who take extreme risks can perpetuate a maladaptive cycle of catastrophic risk tolerance, because a few of them get lucky and exhibit payoffs that are much higher than those of the median cautious individual, even if most individuals who take these risks end up ruined.

3.2.2 Deprivation leads to a population-level pattern of both extreme risk avoidance and extreme risk tolerance

Under conditions of severe deprivation, where wealth buffers are minimal ($\aleph = 0.05$), populations evolve towards a distribution of risk attitudes that is highly skewed (Figure 5). Most individuals converge on extremely low stakes, effectively avoiding nearly all risk, as even a small gamble could lead to ruin. Yet, a minority of individuals—those who learn from individuals with incidental early successes or anomalous circumstances—adopt disproportionately high stakes, demonstrating extreme risk tolerance. This asymmetric pattern emerges because under deprivation, conservative strategies dominate but occasional outliers can still achieve and maintain high-risk behaviors (which they acquire through highly exploratory and/or payoff-biased learning) if they survive early hazards, and whose high growth rates can lead to high cultural evolutionary contributions to the next generation when they occur in an environment dominated by small-growth risk-avoiders. The result is a population composed primarily of extremely cautious risk-avoiders with a conspicuous tail of high-risk-takers (who will also trend to be payoff-biased, as noted above).

3.2.3 Extreme environmental change only undermines conservative learning when wealth buffers are high

In Figure 6, we show the results of repeating our analysis while allowing for extreme environmental change. We allow the environment to oscillate from one generation to the next from a “low edge” in $[0.05, 0.25]$ to a high edge (which we fix at 0.35). We chose these conditions to simulate extreme change: no environmental edge can remain for more than one generation and when it changes, it does so significantly. The most important point in this case is that while in environments with high wealth buffers elder influence decreases due to elder knowledge becoming “obsolete” when the environment changes, it remains at high levels when wealth buffers are low. This means that selection for elder influence remains strong even under extreme environmental change in impoverished populations. This wealth and risk mediation brings more nuance to the theoretical results reported by Deffner

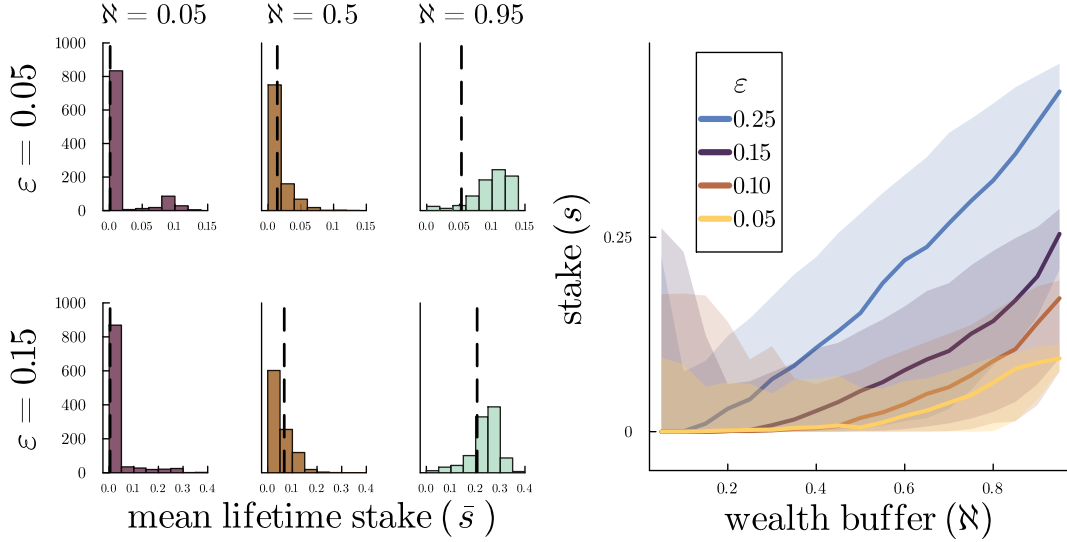


Figure 5: **Deprivation leads to skewed risk-attitude distributions.** The left panels show distributions of mean lifetime stakes ($\bar{s}$) at $T = 2500$ for different combinations of wealth buffers and environmental edges. Vertical dashed lines indicate the approximate Kelly stake given by Equation 12. Under severe deprivation (low $\aleph$), nearly all individuals adopt minuscule stakes, avoiding risk almost entirely, while a small fraction push into high-stake territory, generating a highly-skewed distribution. As conditions improve (greater wealth buffers), stakes shift upward and the skewness subsides, allowing for more moderate, though occasionally risky, strategies. On the right, the shaded areas represent the 5% and 95% quantiles of the stake distributions. The solid lines are the same median stakes as in Figure 4. Note how, under very low $\aleph$, the upper quantile region expands considerably, even when the median stake tends to zero. As $\aleph$ increases, the upper quantiles shrink dramatically and the lower quantiles expand as the median stake increases, as lucky risk-takers have less of a cultural impact when the median individual can experience some growth.

and McElreath (2022), as they show that the use of elder information is still selected for under rapid change when risk of ruin is high. This occurs because it is more important to avoid ruin during difficult periods than it is to grow during favorable times. Peer influence, on the other hand, increases sharply as wealth buffers increase and elder influence is discarded, becoming the main source of social information that wealthy agents use in their favor.

Optimism tends to decline throughout adulthood in deprived and wealthy settings

Figure 7 illustrates how agents adjust their stakes over the course of their adult lifetimes across three different wealth buffer values ($\aleph \in \{0.05, 0.5, 0.95\}$). Each trajectory shows the average proportional change in stake relative to the starting level, providing insight into how agents dynamically modulate their risk-taking as they gain experience. When wealth buffers are minimal ($\aleph = 0.05$), agents rapidly reduce their stakes soon after reaching adulthood. This quick downward adjustment reflects the high perceived danger of ruin in these precarious environments under the consequences of juvenile optimism, prompting quick conservative shifts in adult risk-taking behavior. At intermediate wealth buffers ($\aleph = 0.5$), stakes start off highly pessimistic, and they gradually rise over time. Despite this relative increase, the absolute stakes remain extremely low ($\bar{s} \approx 0.01$), indicating that while agents become somewhat more confident, their actual level of risk-taking is still minimal. In other words, the relative increase is a subtle adjustment occurring at the lowest levels of engagement in risky behavior. By contrast, agents starting with substantial wealth buffers ($\aleph = 0.95$) show almost no change in their stakes throughout adulthood. With a high degree of financial security, there is little incentive to revise their risk attitudes.

3.3 Class inequality and parochialism

To explore the effects of structural socioeconomic heterogeneity, we introduced class structure into the population. Each agent is assigned a class tag $C \in \{c_0, c_1\}$ with probabilities f and $1 - f$, respectively. Classes differ in their wealth buffers and potentially in other environmental parameters, reflecting systematic inequalities in resource access and stability. We also allowed the evolution of parochialism ($\mathcal{P}$), whereby agents preferentially learn from their own class, ignoring out-group members when acquiring risk-related information. Figure 8 shows the evolutionary outcomes of the learning strategies, now given by $\mathbf{L}_i = (\alpha_i, \beta_i, \delta_i, \mathcal{L}_i, \mathcal{P}_i)$, for two classes facing different wealth buffers. The blue lines represent the behavior of the *bourgeoisie* (c_0), while the maroon lines represent the (relatively disadvantaged) *working class* (c_1). We describe the trends in this figure below. We plot results for $f = 0.5$, $\epsilon = 0.25$, $n = 15$, $\tau = 15$, $m = 15$ and $\aleph_{c_0} = 0.95$, and we manipulate inequality by varying the wealth buffer of the working class $\aleph_{c_1}$, expanding to other parameter values in Figure S5. Our simulated data also shows that the exhibited patterns are generally robust to different values of f (see code supplement).

3.3.1 Inequality exacerbates learning differences between classes

In general, in the presence of class-structured populations and structural socioeconomic inequality, the advantaged bourgeoisie evolves a learning strategy characterized by lower elder influence (α), higher reliance on peer information (β), lower sensitivity (δ), neutral parochialism and higher payoff bias as inequality increases. The conditions of greater abundance and stability encourage more optimistic risk-taking, leading to more diversified and insensitive learning strategies. Conversely, the more disadvantaged working class tends toward higher elder influence, higher sensitivity, higher parochialism and lower payoff bias as inequality increases. Reliance on parochial cues limits the integration of out-group information, which might be disadvantageous to utilize given the disparate

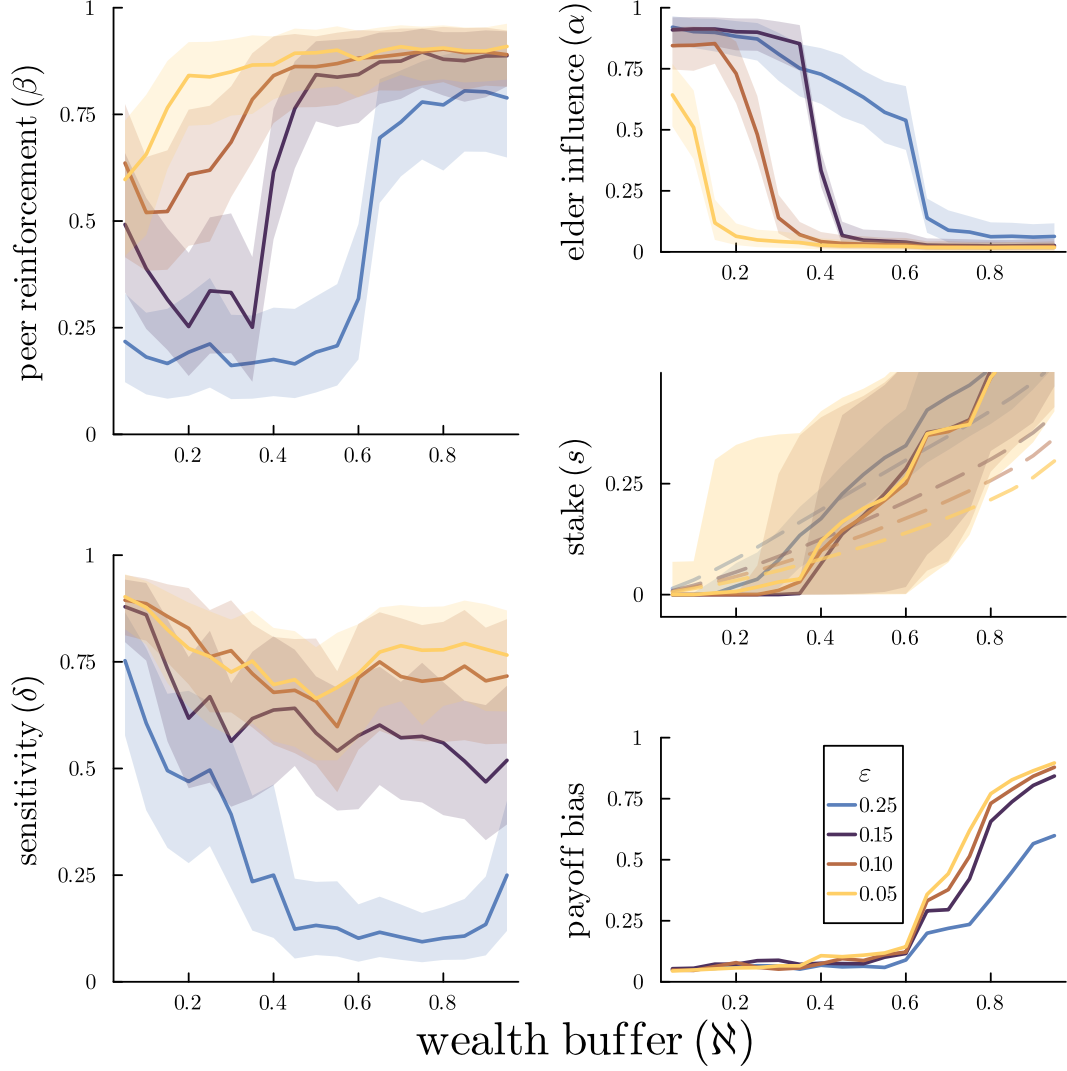


Figure 6: **The effects of environmental change on learning strategies.** In these plots, the environmental edge oscillates between a value (the edge shown in the plot legend) and a high edge of 0.35. This showcases the effects on the evolved learning strategies under extreme conditions of generational environmental change. Even in these conditions, the evolution of elder influence is highly dependent on wealth buffers, with low buffering still leading to the evolution of high elder influence. On the other hand, fast environmental change makes it so that elder influence crashes and peer reinforcement becomes highly favored as wealth buffers increase, a pattern shared by payoff bias. Other than oscillation in environmental edges, the parameters used are the same as in Figure 4.

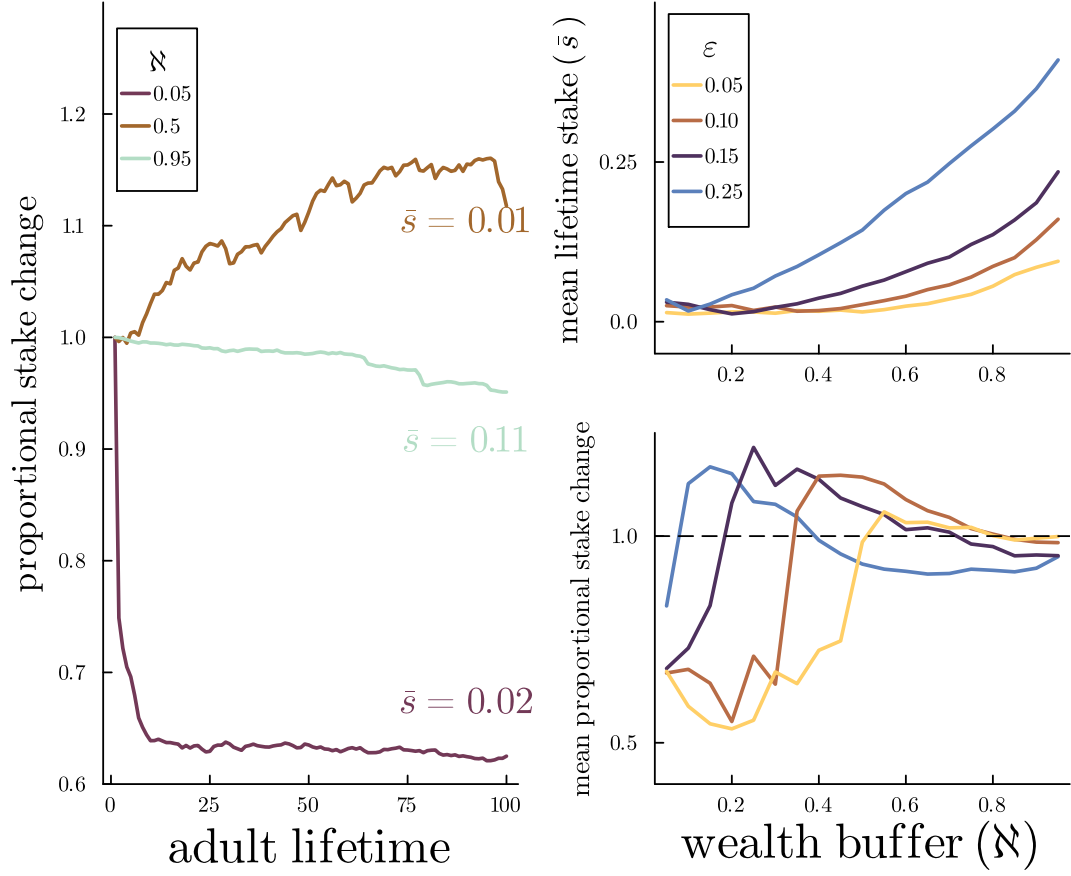


Figure 7: **Average proportional change in stake throughout adulthood relative to initial levels under varying wealth buffers.** **Left:** Each line corresponds to a different wealth buffer ($\aleph$) and shows how the average stake changes as agents progress through their adult lifetime ($T = 100$ periods). Stake trajectories are averaged across $N = 1000$ individuals per condition, and environmental edges are fixed at $\epsilon = 0.05$. The vertical axis indicates the proportion of the starting stake at a particular time period, with 1.0 representing no change. When wealth buffers are small ($\aleph = 0.05$, purple line), agents rapidly reduce their stakes, stabilizing at about 60% of their initial level. Under intermediate buffers ($\aleph = 0.5$, brown line), stakes gradually increase to around 120% of initial levels, reflecting cautious yet adaptive risk-taking. However, average lifetime stakes are the lowest ($\bar{s} \approx 0.01$) at these intermediate wealth buffer values, so these increases are extremely small. With large buffers ($\aleph = 0.95$, green line), stakes remain near their starting values with slight decreases, indicating that high baseline security minimizes the need for lifetime adjustments. Text labels on the right indicate the mean lifetime stakes ($\bar{s}$) corresponding to each scenario. **Right:** mean lifetime stakes and mean proportional stake change across 25 simulations, showing a general view of the above described trend. Default parameters used are the same as in Figure 4.

starting conditions between classes (Smaldino and Pérez Velilla, 2024). The result is a divergence in both risk-taking and learning strategies, driven by the interplay of class-based resource inequalities and selective social learning.

3.3.2 Payoff bias as a form of parochialism for the rich

Parochialism provides an important buffer for the working class, allowing it to avoid maladaptively adopting the riskier practices of out-group members who operate under more forgiving conditions. By contrast, the wealthier bourgeoisie—benefiting from overall safer conditions—maintains a more balanced approach, integrating information from multiple sources without a strong need for selective parochial filtering. However, as the wealth buffers of the working class decrease, increasing the inequality in the population, a higher proportion of wealthy individuals adopt payoff bias as a way to selectively learn from other rich individuals as inequality amplifies. In the working class, payoff bias remains low due to the high risk of ruin it leads to when imitating outlier successes. This difference suggests that within stratified societies, conditions that facilitate consistent wealth accumulation and low-risk environments further incentivize the selective imitation of high performers, but only for the wealthy. Together, these results show how socioeconomic stratification and parochialism can coevolve to produce starkly different risk cultures between social classes.

Curiously enough, the adoption of payoff bias as a parochial strategy by the bourgeois actually leads some of the individuals in this class to end up more risk averse on average throughout their lifetimes than they would be with unbiased learning. This is a consequence of the overly-optimistic effects of payoff bias: once a significant chunk of the bourgeois population is adopting risky behavior associated with high payoffs, some of them inevitably end up experiencing ruin due to catastrophic optimism, inducing pessimistic adjustments on their peers. And, in order to avoid becoming too pessimistic, the bourgeois also evolve a culture of minimal sensitivity to peer ruin.

In Figure S5, we also show that when inequality grows to extreme degrees (for example, when the environmental edge of the bourgeois also exceeds that of the working class), then the bourgeois not only adopt higher payoff bias but also end up adopting explicit parochialism as well, as too much payoff bias can become costly relative to direct parochial biases.

4 Discussion

We developed a cultural evolutionary model to explore how risk attitudes emerge from the interplay of individual life experiences, cultural inheritance, and environmental conditions. Our framework highlights the importance of wealth buffers, environmental edges, and social learning strategies—encompassing juvenile exploration, peer influence, elder influence, payoff bias, and parochialism—in shaping the development and transmission of risk attitudes. By simulating how learning strategies evolve over multiple generations, we have generated insights into how cultural processes can generate variation in risk attitudes within and between populations, as well as across socioeconomic strata.

Conservative vs. exploratory learning in risky environments

Coming from a wealthy family and/or having access to robust societal safety nets allows individuals to take more risk, and thus to achieve higher lifetime growth, while keeping the risk of ruin low. Our results show that when environments pose significant risks of ruin (due to low wealth buffers and/or uncertain payoffs), conservative learning strategies—characterized by high reliance on elder

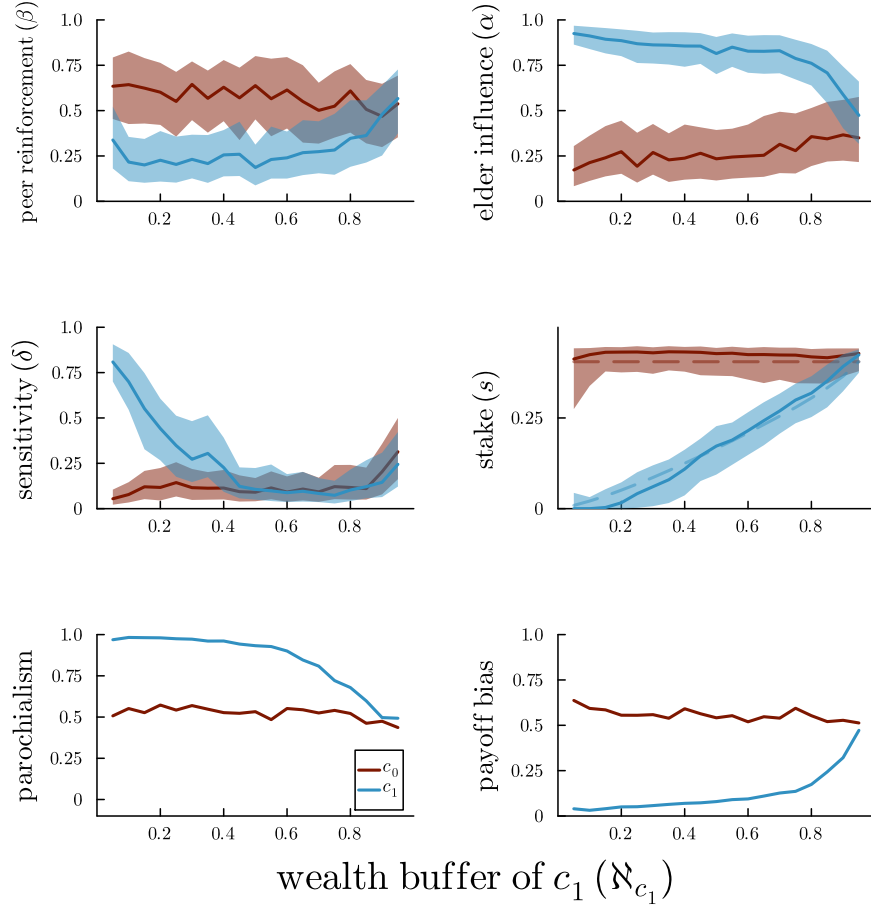


Figure 8: **Effects of class inequality and parochialism on learning strategies and risk attitudes.** Two classes, c_0 (maroon) and c_1 (blue), face distinct wealth buffers. $\aleph_{c_0}$ is fixed at 0.95, while $\aleph_{c_1}$ is varied in the x-axis and treated as a measure of structural inequality. Inequality thus decreases as we move to the right end of the x-axis. The environmental edge is fixed at $\epsilon = 0.25$ for both classes. Each panel shows median evolved values and interquartile ranges of parameters over multiple simulation runs. Peer influence (β) remains generally higher in the bourgeoisie (c_0), while the working class (c_1) relies less on intra-generational knowledge due to its associated risk of catastrophic optimism. Payoff bias differs between classes, remaining lower for the working class due to higher ruin risk, and increasing for the bourgeoisie as inequality increases, due to its value as a parochial strategy when wealth buffers are high. Sensitivity (δ) also shows class-specific trajectories, with the working class increasing their responsiveness to peer ruin as their wealth buffers decrease, while the wealthy become extremely insensitive with the increase in inequality. For elder influence (α) the wealthy rely less on inter-generational influence as inequality increases. Parochialism ($\mathcal{P}$) is more pronounced in the working class, allowing it to avoid adopting out-group strategies ill-suited to its harsher environment, while the bourgeoisie can rely on payoff bias for within-class learning as inequality grows extreme. Due to the adoption of payoff bias, the stakes (s) of the wealthy group decrease slightly as inequality increases, as payoff bias leads to excessive initial risk tolerance and thus to the observed ruin of some individuals.

influence and low payoff bias—are generally favored. These conservative strategies curb overly-optimistic risk-taking and help prevent catastrophic losses. They transmit accumulated knowledge about environmental hazards across generations, producing cultural inertia that can stabilize risk-averse attitudes. Such conservatism is adaptive in the sense that it prioritizes survival and avoids ruin, especially when a single miscalculation can have irreversible consequences.

In particularly deprived conditions, individuals tend to start adulthood with optimistic risk attitudes that quickly crash after they witness their (similarly optimistic) peers experience ruin, leading to the development of high risk avoidance soon after exposure to real-life risks begins. However, groups under extreme deprivation also produce a small but stable proportion of extreme risk-takers, of which some (even fewer) lucky ones will enjoy great relative success, and who proceed to have significant cultural contributions to the next generation. These results mirror the empirical results of de Courson et al. (2024) and provide a mechanistic explanation for them: in poverty, those individuals who take extreme risks tend to fail spectacularly, but a few of them will attain payoffs that are so high relative to other individuals in the population that they manage to culturally influence a small but significant proportion of the next generation. In accordance with what de Courson et al. suggest, this pattern can explain why extremely risky behaviors, such as property crime, are disproportionally observed in impoverished populations, even though most individuals in this category are likely to be extremely risk averse (de Courson et al., 2024).

Conversely, when wealth buffers are large and environmental conditions are more forgiving, agents can afford more exploratory and diversified learning strategies. Here, juvenile risk-free exploration and peer influence become more impactful, and the costs of mistakes diminish. Under these conditions, risk-taking becomes a viable strategy for achieving greater wealth growth, and agents exhibit a higher stability in their risk attitudes throughout their lifetime, with smaller decreases (or even slight increases) throughout (Leonard and Sommerville, 2024). Elder influence remains useful but may reduce in importance as newer generations learn to navigate the environment effectively on their own. Moreover, payoff bias emerges as a “luxury of the rich,” enabling agents in affluent conditions to imitate rare high performers, thereby amplifying success stories and fostering greater tolerance for risk.

Cultural inheritance, early adversity, and generational trauma

Our model clarifies how cultural inheritance interacts with individual-level development to produce changing risk attitudes over the lifespan. Juveniles start off optimistic, benefiting from risk-free exploration and peer learning that shield them from the immediate consequences of over-betting. As they transition into adulthood, observing peers who face ruin leads to increased sensitivity and more cautious, pessimistic risk behavior—especially in harsher environments. This pattern resonates with the idea that early adversity can accelerate the shift from exploratory to exploitative behavior (Frankenhuis and Gopnik, 2023; Gopnik, 2020) and aligns with empirical findings that childhood hardship can result in greater risk aversion later in life (Leonard and Sommerville, 2024). Our model shows that pessimistic adjustment can be adaptive at a population level, even if at an individual level it can lead to cases of over-adjustment, leading to extreme risk avoidance. Sensitivity to experienced or observed ruin can be adaptive at the population level, allowing individuals to pull back overly-optimistic risk attitudes, but it can also lead to some agents becoming “traumatized” (extremely risk averse) due to unfortunate combinations of high sensitivity and witnessing too much ruin. Deprived populations and individuals in highly-adverse environments (such as warzones and/or areas subject to environmental catastrophe) will be more prone to experiencing cases of trauma, as individuals in these conditions need to be highly sensitive to adjust their risk attitudes as quickly as they can to avoid ruin themselves. This makes sensitivity to observed ruin a possible evolutionary mechanism

behind psychological trauma responses, such as post-traumatic stress disorder (Kim and Lee, 2014; Huh et al., 2016; Ruderman et al., 2016; Zefferman and Matthew, 2021; Bayer and Shtudiner, 2024).

The interplay of learning strategies also helps explain why certain groups become trapped in risk-averse equilibria, as the risk aversion acquired throughout the lifetime is passed on to successive generations. A particular generation (or a set of successive generations) experiencing an acute period of considerable adversity may lead to the rapid evolution of “cultural poverty traps” through the transmission of the acquired generational trauma due to exposure to community-wide risks and catastrophe (Moya, 2018; Kim and Lee, 2014; Augsburgers and Elbert, 2017). When deprived of substantial buffers, populations risk frequent ruin whenever they try to act in exploratory ways, and such repeated (possibly generational) exposure to failures can entrench conservative, pessimistic attitudes—an instance of cultural lock-in—making it difficult to seize opportunities for growth even if conditions improve, with effects that might persist for several generations before cultural selection leads to the adoption of more diversified and exploratory learning strategies. By contrast, prosperous conditions that are stable for several generations permit the cultural evolution of more flexible, exploratory strategies, allowing risk attitudes to adjust more readily to improving circumstances, and that might feed into other benefit-inducing processes, such as innovation, that we have not explicitly modeled in this paper.

The rise of socioeconomic inequality, parochialism, and class-based risk cultures

Introducing class inequality into our model shows how structural differences in wealth buffers and risk landscapes generate divergent learning strategies and risk cultures. Disadvantaged classes facing harsher conditions and more frequent ruin tend to evolve higher elder influence, even greater sensitivity, and strong parochialism as a strategy to avoid catastrophic optimism. By focusing on intra-class learning and cautious elder wisdom, disadvantaged groups protect themselves from adopting maladaptive, risky strategies that might work only for those with more abundant resources.

In contrast, wealthy classes, operating in safer environments, can tolerate greater risk-taking and integrate more diverse sources of information, including using payoff-biased strategies to avoid learning from the poor when inequality becomes extreme. Over time, this divergence produces stark differences in risk cultures between classes. Whereas the disadvantaged class remains trapped in a conservative, low-stake equilibrium, the affluent class can adopt moderate or even high stakes, using both payoff bias and less parochial learning to adapt flexibly. These findings align with the notion that socioeconomic disparities can perpetuate through cultural channels, as wealthier groups accumulate and transmit risk-taking strategies that lead to continued growth, while poorer groups remain stuck in risk-averse equilibria that limit upward mobility.

Previous work has shown that the Pareto distribution of wealth usually observed empirically can be explained by differential luck, and not so much through differential investment ability (Levy, 2003). This means the rich are not necessarily smarter, they might have just been historically luckier. Our results complement this view: in our simulations, significant wealth advantages arise purely out of chance-based “lucky streaks” and, among those who show excessive risk tolerance, most will experience ruin but a lucky few will experience incredible gains. Adding to this, our model shows that it is possible for these emergent inequalities to become exacerbated by promoting conservative learning strategies for disadvantaged groups, leading to an exploitation-first mindset that might not be receptive to new solutions. Advantaged groups, those who are well-buffered (and many of whom will have inherited the product of their predecessors’ lucky streaks), are more free to make their learning strategies more exploratory, which may grant tangible advantages when it comes to the adoption of cultural innovations, and which can then lead to even more compounding benefits down

the line and faster increases in between-group inequalities. In other words, those who were initially lucky not only obtain higher wealth, but also reap the generational benefits of cultural exploration with large margins of error. The downtrodden, on the other hand, must be extremely cautious about their choices and the behaviors they adopt from others, limiting their exploratory options and locking them into tried-and-true solutions. Only those who put themselves at a significant risk have a small chance to grow their capital, in return for a great chance of becoming ruined.

Connections to existing cultural and psychological frameworks

Our approach provides a plausible way to unify multiple dimensions of cultural variation discussed in the social science literature—such as cultural conservatism (Morin, 2022), tightness-looseness (Gelfand et al., 2011), and WEIRDness (Henrich et al., 2010)—under a broad theoretical umbrella, through their relationship to environmental risk, wealth buffers and cultural-evolutionary dynamics. Thus, differences that have been described as distinct cultural phenomena could be reinterpreted as manifestations of the same underlying adaptive dynamics responding to risk and scarcity. In high-risk, resource-scarce contexts, conservative and parochial learning strategies can lead to the emergence of tight cultural norms as adaptive responses to mitigate ruin. In low-risk, resource-rich environments, exploratory and payoff-biased strategies that lead to loose norms and greater individualism may thrive because the consequences of failure are less dire. Moreover, these differences can happen (and be exacerbated) within populations due to the effects of class stratification.

Our framework also suggests that changes in economic conditions, welfare policies, or social institutions that effectively alter wealth buffers can precipitate cultural shifts in risk tolerance through the incentives they induce on learning strategies, although the presence of large inequalities can mitigate these shifts. However, when these do shifts happen, they can in turn influence patterns of innovation, social trust, and institutional stability, potentially linking micro-level developmental processes to macro-level cultural patterns. If risk attitudes are truly the product of the learning processes we have explored here, then interventions aiming to reduce risk and inequality will have to factor in the cultural inertia of evolved conservative learning strategies, especially for groups that have lived in conditions of deprivation and inequality for several generations. This cultural inertia of conservative learning can look like a “poverty trap” when changes in learning and risk attitudes are expected to happen shortly after the implementation of an intervention, instead of on a longer, generational timescale. Our results also imply that interventions can lead to faster change when they not only focus on improving wealth buffers, but also accompany these improvements with strong encouragement for diversification of learning strategies.

A detailed examination of how our current theoretical work can be used as a way to unify these separate strands of theoretical, empirical and applied research will be necessary as a future direction of this line of research. We have provided, through our theoretical results, the tools to build hypotheses that can kickstart the process of empirical falsification.

5 Conclusion

Our model highlights the importance of cultural evolutionary dynamics in shaping risk attitudes across diverse social and economic landscapes. By showing how individual development, cultural transmission, and environmental conditions interact to produce stable—but malleable—risk strategies, we provide a framework for understanding human risk-taking behavior and its cultural variation over both developmental and generational time. In doing so, we offer new insights into how cultural inertia, social inequality, and environmental uncertainty can perpetuate or transform the risk attitudes that influence economic decisions, resource distribution, and social cohesion.

All lifeforms on Earth have to navigate risk throughout their lifetimes, and must do so through uncertainty reduction strategies. Human individuals and populations, while complex, are not exempt from these challenges. In our view, we produce culture as a way to regularize our environments and make life less volatile, and we allow ourselves to explore once the existential threat of deprivation is no longer weighting heavily in our minds.

References

- Alger, I. (2023). Evolutionarily stable preferences. *Philosophical Transactions of the Royal Society B*, 378(1876):20210505.
- Amir, D., Jordan, M. R., McAuliffe, K., Valeggia, C. R., Sugiyama, L. S., Bribiescas, R. G., Snodgrass, J. J., and Dunham, Y. (2020). The developmental origins of risk and time preferences across diverse societies. *Journal of Experimental Psychology: General*, 149(4):650.
- Asmussen, S. and Albrecher, H. (2010). *Ruin probabilities*, volume 14. World scientific.
- Augsburger, M. and Elbert, T. (2017). When do traumatic experiences alter risk-taking behavior? a machine learning analysis of reports from refugees. *PLoS one*, 12(5):e0177617.
- Baldini, R. (2012). Success-biased social learning: cultural and evolutionary dynamics. *Theoretical population biology*, 82(3):222–228.
- Bayer, Y. M. and Shtudiner, Z. (2024). Sirens of stress: Financial risk, time preferences, and post-traumatic stress disorder: Evidence from the israel-hamas conflict. *Journal of health psychology*, 29(13):1489–1502.
- Bernoulli, D. (1954). Exposition of a new theory on the measurement. *Econometrica*, 22(1):23–36.
- Bezanson, J., Karpinski, S., Shah, V. B., and Edelman, A. (2012). Julia: A fast dynamic language for technical computing. *arXiv preprint arXiv:1209.5145*.
- Bowles, S. (1998). Endogenous preferences: The cultural consequences of markets and other economic institutions. *Journal of economic literature*, 36(1):75–111.
- Boyd, R. and Richerson, P. J. (1985). *Culture and the evolutionary process*. University of Chicago press.
- Boyd, R., Richerson, P. J., and Henrich, J. (2011). The cultural niche: Why social learning is essential for human adaptation. *Proceedings of the National Academy of Sciences*, 108(supplement_2):10918–10925.
- Burghardt, G. M., Pellis, S. M., Schank, J. C., Smaldino, P. E., Vanderschuren, L. J., and Palagi, E. (2024). Animal play and evolution: Seven timely research questions about enigmatic phenomena. *Neuroscience & Biobehavioral Reviews*, 160:105617.
- Cavalli-Sforza, L. L. and Feldman, M. W. (1981). *Cultural transmission and evolution: A quantitative approach*. Number 16. Princeton University Press.
- Chudek, M. and Henrich, J. (2011). Culture–gene coevolution, norm-psychology and the emergence of human prosociality. *Trends in Cognitive Sciences*, 15(5):218–226.

- Datseris, G., Vahdati, A. R., and DuBois, T. C. (2022). Agents. jl: a performant and feature-full agent-based modeling software of minimal code complexity. *Simulation*, page 00375497211068820.
- de Courson, B., Frankenhuis, W., and Nettle, D. (2024). Poverty is associated with both risk avoidance and risk taking: an empirical test of the desperation threshold model. *SocArXiv*.
- Deffner, D. and McElreath, R. (2022). When does selection favor learning from the old? social learning in age-structured populations. *PloS one*, 17(4):e0267204.
- Frankenhuis, W. E. and Gopnik, A. (2023). Early adversity and the development of explore–exploit tradeoffs. *Trends in Cognitive Sciences*.
- Gelfand, M. J., Gavrillets, S., and Nunn, N. (2024). Norm dynamics: Interdisciplinary perspectives on social norm emergence, persistence, and change. *Annual Review of Psychology*, 75(1):341–378.
- Gelfand, M. J., Raver, J. L., Nishii, L., Leslie, L. M., Lun, J., Lim, B. C., Duan, L., Almaliach, A., Ang, S., Arnadottir, J., et al. (2011). Differences between tight and loose cultures: A 33-nation study. *science*, 332(6033):1100–1104.
- Gillespie, J. H. (1974). Natural selection for within-generation variance in offspring number. *Genetics*, 76(3):601–606.
- Gopnik, A. (2020). Childhood as a solution to explore–exploit tensions. *Philosophical Transactions of the Royal Society B*, 375(1803):20190502.
- Harrison, G. W. and Rutström, E. (2008). Risk aversion in the laboratory. In *Risk aversion in experiments*, pages 41–196. Emerald Group Publishing Limited.
- Henrich, J., Heine, S. J., and Norenzayan, A. (2010). The weirdest people in the world? *Behavioral and brain sciences*, 33(2-3):61–83.
- Huh, H. J., Baek, K., Kwon, J.-H., Jeong, J., and Chae, J.-H. (2016). Impact of childhood trauma and cognitive emotion regulation strategies on risk-averse and loss-averse patterns of decision-making in patients with depression. *Cognitive neuropsychiatry*, 21(6):447–461.
- JE, O. and CT, T. (2021). Reactive, reproducible, collaborative: computational notebooks evolve. *Nature*, 593.
- Kandler, A., Fogarty, L., and Karsdorp, F. (2023). The interplay between age structure and cultural transmission. *PLOS Computational Biology*, 19(7):e1011297.
- Kelly, J. L. (1956). A new interpretation of information rate. *The Bell System Technical Journal*, 35(4):917–926.
- Kendal, R. L., Boogert, N. J., Rendell, L., Laland, K. N., Webster, M., and Jones, P. L. (2018). Social learning strategies: Bridge-building between fields. *Trends in cognitive sciences*, 22(7):651–665.
- Kim, Y.-I. and Lee, J. (2014). The long-run impact of a traumatic experience on risk aversion. *Journal of Economic Behavior & Organization*, 108:174–186.
- Knight, F. H. (1921). *Risk, uncertainty and profit*. Hart, Schaffner and Marx.
- Laland, K. N. (2004). Social learning strategies. *Animal Learning & Behavior*, 32(1):4–14.

- Leonard, J. A. and Sommerville, J. A. (2024). A unified account of why optimism declines in childhood. *Nature Reviews Psychology*, pages 1–14.
- Levy, H. and Levy, M. (2009). The safety first expected utility model: Experimental evidence and economic implications. *Journal of Banking & Finance*, 33(8):1494–1506.
- Levy, M. (2003). Are rich people smarter? *Journal of Economic theory*, 110(1):42–64.
- Levy, M. (2010). Evolution of risk aversion: The ‘having descendants forever’ approach. *Available at SSRN 1688265*.
- Levy, M. (2024). Relative risk aversion must be close to 1. *Annals of Operations Research*, pages 1–9.
- Mallpress, D. E., Fawcett, T. W., Houston, A. I., and McNamara, J. M. (2015). Risk attitudes in a changing environment: An evolutionary model of the fourfold pattern of risk preferences. *Psychological Review*, 122(2):364.
- McElreath, R., Wallin, A., and Fasolo, B. (2013). The evolutionary rationality of social learning. In Hertwig, R., Hoffrage, U., and ABC Research Group, editors, *Simple heuristics in a social world*, pages 381–408. Oxford University Press.
- Miller, P., Podvysotska, T., Betancur, L., and Votruba-Drzal, E. (2021). Wealth and child development: differences in associations by family income and developmental stage. *RSF: The Russell Sage Foundation Journal of the Social Sciences*, 7(3):154–174.
- Morin, O. (2022). Cultural conservatism. *Journal of Cognition and Culture*, 22(5):406–420.
- Moya, A. (2018). Violence, psychological trauma, and risk attitudes: Evidence from victims of violence in colombia. *Journal of Development Economics*, 131:15–27.
- Mulder, M. B., Bowles, S., Hertz, T., Bell, A., Beise, J., Clark, G., Fazzio, I., Gurven, M., Hill, K., Hooper, P. L., et al. (2009). Intergenerational wealth transmission and the dynamics of inequality in small-scale societies. *science*, 326(5953):682–688.
- Nettle, D. and Frankenhuys, W. E. (2020). Life-history theory in psychology and evolutionary biology: one research programme or two? *Philosophical Transactions of the Royal Society B*, 375(1803):20190490.
- Netzer, N., Robson, A. J., Steiner, J., and Kocourek, P. (2022). Endogenous risk attitudes. *CESifo Working Papers*, (9547).
- Paulsen, D. J., Platt, M. L., Huettel, S. A., and Brannon, E. M. (2012). From risk-seeking to risk-averse: the development of economic risk preference from childhood to adulthood. *Frontiers in psychology*, 3:313.
- Poundstone, W. (2010). *Fortune’s formula: The untold story of the scientific betting system that beat the casinos and Wall Street*. Hill and Wang.
- Robson, A. and Samuelson, L. (2022). The evolution of risk attitudes with fertility thresholds. *Journal of Economic Theory*, 205:105552.
- Robson, A. J. and Orr, H. A. (2021). Evolved attitudes to risk and the demand for equity. *Proceedings of the National Academy of Sciences*, 118(26):e2015569118.

- Robson, A. J. and Samuelson, L. (2019). Evolved attitudes to idiosyncratic and aggregate risk in age-structured populations. *Journal of Economic Theory*, 181:44–81.
- Rodems, R. and Pfeffer, F. T. (2021). Avoiding material hardship: The buffer function of wealth. *Journal of European Social Policy*, 31(5):517–532.
- Rogers, A. R. (1988). Does biology constrain culture? *American anthropologist*, 90(4):819–831.
- Ruderman, L., Ehrlich, D. B., Roy, A., Pietrzak, R. H., Harpaz-Rotem, I., and Levy, I. (2016). Posttraumatic stress symptoms and aversion to ambiguous losses in combat veterans. *Depression and anxiety*, 33(7):606–613.
- Schreiber, S. J. (2015). Unifying within-and between-generation bet-hedging theories: an ode to jh gillespie. *The American Naturalist*, 186(6):792–796.
- Smaldino, P. E., Palagi, E., Burghardt, G. M., and Pellis, S. M. (2019). The evolution of two types of play. *Behavioral Ecology*, 30(5):1388–1397.
- Smaldino, P. E. and Pérez Velilla, A. (2024). The evolution of similarity-biased social learning. *SocArXiv*.
- Smith, D. M., Langa, K. M., Kabeto, M. U., and Ubel, P. A. (2005). Health, wealth, and happiness: Financial resources buffer subjective well-being after the onset of a disability. *Psychological science*, 16(9):663–666.
- Strauss, E. D. and Shizuka, D. (2022). The ecology of wealth inequality in animal societies. *Proceedings of the Royal Society B*, 289(1974):20220500.
- Turner, M. A., Moya, C., Smaldino, P. E., and Jones, J. H. (2023). The form of uncertainty affects selection for social learning. *Evolutionary Human Sciences*, 5:e20.
- Van Tilburg, W. A. and Mahadevan, N. (2020). When imitating successful others fails: Accidentally successful exemplars inspire risky decisions and can hamper performance. *Quarterly Journal of Experimental Psychology*, 73(6):941–956.
- Weber, E. U. and Hsee, C. K. (1999). Models and mosaics: Investigating cross-cultural differences in risk perception and risk preference. *Psychonomic bulletin & review*, 6(4):611–617.
- Xu, Y., Harms, M. B., Green, C. S., Wilson, R. C., and Pollak, S. D. (2023). Childhood unpredictability and the development of exploration. *Proceedings of the National Academy of Sciences*, 120(49):e2303869120.
- Zefferman, M. R. and Matthew, S. (2021). Combat stress in a small-scale society suggests divergent evolutionary roots for posttraumatic stress disorder symptoms. *Proceedings of the National Academy of Sciences*, 118(15):e2020430118.
- Zhang, R., Brennan, T. J., and Lo, A. W. (2014). The origin of risk aversion. *Proceedings of the National Academy of Sciences*, 111(50):17777–17782.